

**EXAMPLE 3** Solve the following equations:

(i)  $3x + 17 \leq 2(1 - x)$

(ii)  $2(2x + 3) - 10 \leq 6(x - 2)$

**SOLUTION** (i) We have,

$$3x + 17 \leq 2(1 - x)$$

$$\Rightarrow 3x + 17 \leq 2 - 2x$$

$$\Rightarrow 3x + 2x \leq 2 - 17$$

[Transposing  $-2x$  to LHS and 17 to RHS]

$$\Rightarrow 5x \leq -15$$

$$\Rightarrow \frac{5x}{5} \leq \frac{-15}{5}$$

$$\Rightarrow x \leq -3$$

$$\Rightarrow x \in (-\infty, -3]$$

Hence, the solution set of the given inequation is  $(-\infty, -3]$ , which can be graphed on real line as shown in Fig. 15.6.

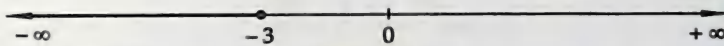


Fig. 15.6

(ii) We have,

$$2(2x + 3) - 10 \leq 6(x - 2)$$

$$\Rightarrow 4x + 6 - 10 \leq 6x - 12$$

$$\Rightarrow 4x - 4 \leq 6x - 12$$

$$\Rightarrow 4x - 6x \leq -12 + 4$$

[Transposing  $-4$  to RHS and  $6x$  to LHS]

$$\Rightarrow -2x \leq -8$$

$$\Rightarrow \frac{-2x}{-2} \geq \frac{-8}{-2}$$

$$\Rightarrow x \geq 4$$

$$\Rightarrow x \in [4, \infty)$$

Hence, the solution set of the given inequation is  $[4, \infty)$  which can be graphed on real line as shown in Fig. 15.7.

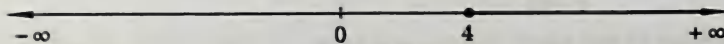


Fig. 15.7

**EXAMPLE 4** Solve the following inequations:

(i)  $\frac{2x - 3}{4} + 9 \geq 3 + \frac{4x}{3}$

(ii)  $\frac{5x - 2}{3} - \frac{7x - 3}{5} > \frac{x}{4}$

(iii)  $\frac{1}{2} \left( \frac{3}{5}x + 4 \right) \geq \frac{1}{3}(x - 6)$

(iv)  $\frac{3(x - 2)}{5} \geq \frac{5(2 - x)}{3}$

**SOLUTION** (i) We have,

$$\frac{2x - 3}{4} + 9 \geq 3 + \frac{4x}{3}$$

$$\Rightarrow \frac{2x - 3}{4} - \frac{4x}{3} \geq 3 - 9$$

[Transposing  $\frac{4x}{3}$  to LHS and 9 to RHS]

$$\Rightarrow \frac{3(2x - 3) - 16x}{12} \geq -6$$

$$\Rightarrow \frac{6x - 9 - 16x}{12} \geq -6$$

$$\Rightarrow \frac{-9 - 10x}{12} \geq -6$$

$$\Rightarrow -9 - 10x \geq -72$$

[Multiplying both sides by 12]

$$\Rightarrow -10x \geq -72 + 9$$

$$\Rightarrow -10x \geq -63$$

$$\Rightarrow \frac{-10x}{-10} \leq \frac{-63}{-10}$$

$$\Rightarrow x \leq \frac{63}{10}$$

$$\Rightarrow x \in (-\infty, 63/10]$$

Hence, the solution set of the given inequation is  $(-\infty, 63/10]$ . This can be graphed on real line as shown in Fig. 15.8.



Fig. 15.8

(ii) We have,

$$\frac{5x-2}{3} - \frac{7x-3}{5} > \frac{x}{4}$$

$$\Rightarrow \frac{5(5x-2) - 3(7x-3)}{15} > \frac{x}{4}$$

$$\Rightarrow \frac{25x-10-21x+9}{15} > \frac{x}{4}$$

$$\Rightarrow \frac{4x-1}{15} > \frac{x}{4}$$

$$\Rightarrow 4(4x-1) > 15x$$

[Multiplying both sides by 60 i.e. lcm of 15 and 4]

$$\Rightarrow 16x - 4 > 15x$$

$$\Rightarrow 16x - 15x > 4$$

[Transposing 15x to LHS and -4 to RHS]

$$\Rightarrow x > 4$$

$$\Rightarrow x \in (4, \infty)$$

Hence, the solution set of the given inequation is  $(4, \infty)$ . This can be graphed on the real line as shown in Fig. 15.9.



Fig. 15.9

(iii) We have,

$$\frac{1}{2} \left( \frac{3}{5}x + 4 \right) \geq \frac{1}{3}(x-6)$$

$$\Rightarrow \frac{1}{2} \left( \frac{3x+20}{5} \right) \geq \frac{1}{3}(x-6)$$

$$\Rightarrow \frac{3x+20}{10} \geq \frac{x-6}{3}$$

$$\Rightarrow 3(3x+20) \geq 10(x-6)$$

[Multiplying both sides by 30 i.e. the lcm of 3 and 10]

$$\Rightarrow 9x + 60 \geq 10x - 60$$

$$\Rightarrow 9x - 10x \geq -60 - 60$$

[Transposing 10x on LHS and 60 on RHS]

$$\Rightarrow -x \geq -120$$

$$\Rightarrow x \leq 120$$

[ Multiplying both sides by  $-1$ ]

$$\Rightarrow x \in (-\infty, 120]$$

Hence, the solution set of the given in equation is  $(-\infty, 120]$  which can be graphed on real line as shown in Fig. 15.10.



Fig. 15.10

(iv) We have,

$$\frac{3(x-2)}{5} \geq \frac{5(2-x)}{3}$$

$$\Rightarrow \frac{3x-6}{5} \geq \frac{10-5x}{3}$$

$$\Rightarrow 3(3x-6) \geq 5(10-5x)$$

[Multiplying both sides by 15 i.e. the lcm of 5 and 3]

$$\Rightarrow 9x-18 \geq 50-25x$$

$$\Rightarrow 9x+25x \geq 50+18$$

[Transposing  $-25x$  to LHS and 18 to RHS]

$$\Rightarrow 34x \geq 68$$

$$\Rightarrow \frac{34x}{34} \geq \frac{68}{34}$$

$$\Rightarrow x \geq 2$$

$$\Rightarrow x \in [2, \infty)$$

Hence,  $[2, \infty)$  is the solution set of the given inequation. This solution set can be graphed on real line as shown in Fig. 15.11.

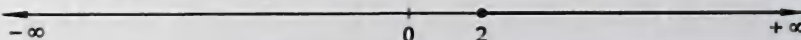


Fig. 15.11

**EXAMPLE 5** Solve the following inequations:

$$(i) \frac{1}{x-2} < 0$$

$$(ii) \frac{x+1}{x+2} \geq 1$$

**SOLUTION** (i) We have,

$$\frac{1}{x-2} < 0$$

$$\Rightarrow x-2 < 0$$

$$\left[ \because \frac{a}{b} < 0 \text{ and } a > 0 \Rightarrow b < 0 \right]$$

$$\Rightarrow x < 2$$

$$\Rightarrow x \in (-\infty, 2)$$

Hence, the solution set of the given inequation is  $(-\infty, 2)$ .

(ii) We have,

$$\frac{x+1}{x+2} \geq 1$$

$$\Rightarrow \frac{x+1}{x+2} - 1 \geq 0$$

$$\Rightarrow \frac{x+1-x-2}{x+2} \geq 0$$

$$\Rightarrow \frac{-1}{x+2} \geq 0$$

$$\Rightarrow x+2 < 0$$

$$\Rightarrow x < -2$$

$$\Rightarrow x \in (-\infty, -2)$$

$$\left[ \because \frac{a}{b} > 0 \text{ and } a < 0 \Rightarrow b < 0 \right]$$

Hence, the solution set of the given inequation is  $(-\infty, -2)$ .

### Type II EQUATIONS OF THE FORM

$$\frac{ax+b}{cx+d} > k, \text{ or } \frac{ax+b}{cx+d} \geq k, \text{ or } \frac{ax+b}{cx+d} < k, \text{ or } \frac{ax+b}{cx+d} \leq k$$

In order to solve this type of inequation, we use the following algorithm.

#### ALGORITHM

**STEP I** Obtain the inequation.

**STEP II** Transpose all terms on LHS.

**STEP III** Simplify LHS of the inequation obtained in step II to obtain an inequation of the form

$$\frac{px+q}{rx+s} > 0, \text{ or } \frac{px+q}{rx+s} \geq 0, \text{ or } \frac{px+q}{rx+s} < 0, \text{ or } \frac{px+q}{rx+s} \leq 0.$$

**STEP IV** Make coefficient  $x$  positive in numerator and denominator if they are not.

**STEP V** Equate numerator and denominator separately to zero and obtain the values of  $x$ . These values of  $x$  are generally called critical points.

**STEP VI** Plot the critical points obtained in step V on real line. These points will divide the real line in three regions.

**STEP VII** In the right most region the expression on LHS of the inequation obtained in step IV will be positive and in other regions it will be alternatively negative and positive. So, mark positive sign in the right most region and then mark alternatively negative and positive signs in other regions.

**STEP VIII** Select appropriate region on the basis of the sign of the inequation obtained in step IV. Write these regions in the form of intervals to obtain the desired solution sets of the given inequation.

**EXAMPLE 6** Solve the following linear inequations:

$$(i) \frac{x-3}{x-5} > 0$$

$$(ii) \frac{x-2}{x+5} > 2$$

**SOLUTION** (i) We have,

$$\frac{x-3}{x-5} > 0$$

...(i)

Equating  $x-3$  and  $x-5$  to zero, we obtain  $x = 3, 5$  as critical points. Plot these points on real line as shown in Fig. 15.12. The real line is divided into three regions. In the right most region the expression on LHS of (i) is positive and in the remaining two regions it is alternatively negative and positive as shown in Fig. 15.12.

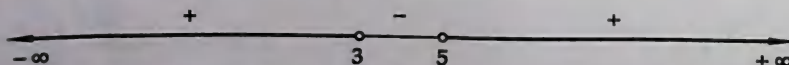


Fig. 15.12

Since the expression in (i) is positive, so the solution set of the given inequation is the union of regions containing positive signs. Hence, from Fig. 15.12

$$\frac{x-3}{x-5} > 0 \Rightarrow x \in (-\infty, 3) \cup (5, \infty)$$

Hence, the solution set of the given inequation is  $(-\infty, 3) \cup (5, \infty)$  as shown in Fig. 15.12.

(ii) We have,

$$\begin{aligned} & \frac{x-2}{x+5} > 2 \\ \Rightarrow & \frac{x-2}{x+5} - 2 > 0 \\ \Rightarrow & \frac{x-2-2(x+5)}{x+5} > 0 \\ \Rightarrow & \frac{x-2-2x-10}{x+5} > 0 \\ \Rightarrow & \frac{-x-12}{x+5} > 0 \\ \Rightarrow & \frac{x+12}{x+5} < 0 \end{aligned} \quad \left[ \begin{array}{l} \text{Multiplying by } -1 \text{ to make coefficient of} \\ x \text{ positive in the expression in numerator} \end{array} \right] \dots(i)$$

On equating  $x + 12$  and  $x + 5$  to zero, we obtain  $x = -12, -5$  as critical points. These points are plotted on number line as shown in Fig. 15.12. The real line is divided into three regions and the signs of LHS of inequation (i) are marked. Since the inequation in (i) possesses less than sign which means that LHS of the inequation is negative. So, the solution set of the given inequation is the union of the regions containing negative sign in Fig. 15.13. Hence, the solution set of the given inequation is  $(-12, -5)$ .

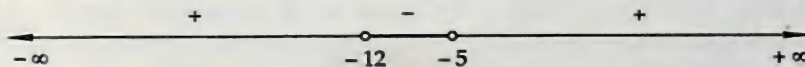


Fig. 15.13

**EXAMPLE 7** Solve the following inequations:

(i)  $\frac{2x+4}{x-1} \geq 5$

(ii)  $\frac{x+3}{x-2} \leq 2$

**SOLUTION** (i) We have,

$$\begin{aligned} & \frac{2x+4}{x-1} \geq 5 \\ \Rightarrow & \frac{2x+4}{x-1} - 5 \geq 0 \\ \Rightarrow & \frac{2x+4-5(x-1)}{x-1} \geq 0 \\ \Rightarrow & \frac{2x+4-5x+5}{x-1} \geq 0 \\ \Rightarrow & \frac{-3x+9}{x-1} \geq 0 \\ \Rightarrow & \frac{3x-9}{x-1} \leq 0 \quad \left[ \text{Multiplying both sides by } -1 \right] \\ \Rightarrow & \frac{3(x-3)}{(x-1)} \leq 0 \\ \Rightarrow & \frac{x-3}{x-1} \leq 0 \quad \left[ \text{Dividing both sides by } 3 \right] \\ \Rightarrow & 1 < x \leq 3 \quad \left[ \text{See Fig. 15.14} \right] \end{aligned}$$

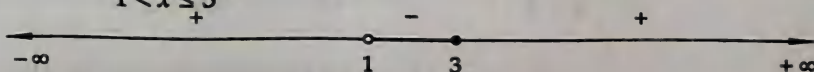


Fig. 15.14

$$\Rightarrow x \in (1, 3]$$

Hence, the solution set of the given inequation is  $(1, 3]$ .

(ii) We have,

$$\frac{x+3}{x-2} \leq 2$$

$$\Rightarrow \frac{x+3}{x-2} - 2 \leq 0$$

$$\Rightarrow \frac{x+3-2x+4}{x-2} \leq 0$$

$$\Rightarrow \frac{-x+7}{x-2} \leq 0$$

$$\Rightarrow \frac{x-7}{x-2} \geq 0$$

[Multiplying both sides by  $-1$ ]

$$\Rightarrow x \in (-\infty, 2) \cup [7, \infty)$$

[See Fig. 15.15]



Fig. 15.15

Hence, the solution set of the given inequation is  $(-\infty, 2) \cup [7, \infty)$ .

### EXERCISE 15.1

#### LEVEL-1

Solve the following linear inequations in  $R$ .

- Solve:  $12x < 50$ , when
  - $x \in R$
  - $x \in Z$
  - $x \in N$
- Solve:  $-4x > 30$ , when
  - $x \in R$
  - $x \in Z$
  - $x \in N$
- Solve:  $4x - 2 < 8$ , when
  - $x \in R$
  - $x \in Z$
  - $x \in N$
- $3x - 7 > x + 1$
- $x + 5 > 4x - 10$
- $2(3 - x) \geq \frac{x}{5} + 4$
- $\frac{3x-2}{5} \leq \frac{4x-3}{2}$
- $\frac{3x+9}{2} \geq -x + 19$
- $\frac{x}{5} < \frac{3x-2}{4} - \frac{5x-3}{5}$
- $\frac{2(x-1)}{5} \leq \frac{3(2+x)}{7}$
- $-(x-3) + 4 < 5 - 2x$
- $\frac{x-1}{3} + 4 < \frac{x-5}{5} - 2$
- $\frac{2x+3}{4} - 3 < \frac{x-4}{3} - 2$
- $\frac{5x}{2} + \frac{3x}{4} \geq \frac{39}{4}$
- $\frac{4+2x}{3} \geq \frac{x}{2} - 3$
- $\frac{2x+3}{5} - 2 < \frac{3(x-2)}{5}$
- $\frac{5-2x}{3} < \frac{x}{6} - 5$
- $x - 2 \leq \frac{5x+8}{3}$
- $\frac{6x-5}{4x+1} < 0$
- $\frac{2x-3}{3x-7} > 0$
- $\frac{3}{x-2} < 1$
- $\frac{1}{x-1} \leq 2$
- $\frac{4x+3}{2x-5} < 6$
- $\frac{5x-6}{x+6} < 1$
- $\frac{5x+8}{4-x} < 2$
- $\frac{x-1}{x+3} > 2$
- $\frac{7x-5}{8x+3} > 4$
- $\frac{x}{x-5} > \frac{1}{2}$

**ANSWERS**

- |                                       |                                            |                                     |
|---------------------------------------|--------------------------------------------|-------------------------------------|
| 1. (i) $(-\infty, 25/6)$              | (ii) $\{\dots -3, -2, -1, 0, 1, 2, 3, 4\}$ | (iii) $\{1, 2, 3, 4\}$              |
| 2. (ii) $(-\infty, -15/2)$            | (ii) $\{\dots, -9, -8\}$                   | (iii) $\phi$                        |
| 3. (i) $(-\infty, 5/2)$               | (ii) $\{\dots, -2, -1, 0, 1, 2\}$          | (iii) $\{1, 2\}$                    |
| 4. $(4, \infty)$                      | 5. $(-\infty, 5)$                          | 6. $[5/2, \infty)$                  |
| 7. $(-\infty, 10/11]$                 | 8. $[11/14, \infty)$                       | 9. $(-\infty, -2)$                  |
| 10. $(-\infty, 2/9)$                  | 11. $[-44, \infty)$                        | 12. $[3, \infty)$                   |
| 13. $(-\infty, -50)$                  | 14. $(-\infty, -13/2)$                     | 15. $(8, \infty)$                   |
| 16. $[-26, \infty)$                   | 17. $(-1, \infty)$                         | 18. $[-7, \infty)$                  |
| 19. $(-1/4, 5/6)$                     | 20. $(-\infty, 3/2) \cup (7/3, \infty)$    | 21. $(-\infty, 2) \cup (5, \infty)$ |
| 22. $(-\infty, 1) \cup [3/2, \infty)$ | 23. $(-\infty, 5/2) \cup (33/8, \infty)$   | 24. $(-6, 3)$                       |
| 25. $(-\infty, 0) \cup (4, \infty)$   | 26. $(-7, -3)$                             | 27. $(-17/25, -3/8)$                |
| 28. $(-\infty, -5) \cup (5, \infty)$  |                                            |                                     |

### 15.5 SOLUTION OF SYSTEM OF LINEAR INEQUATIONS IN ONE VARIABLE

In the previous section, we have learnt how to solve a linear inequation in one variable. In this section, we shall use it to solve a system of linear inequations in one variable. Recall that the solution set of a linear inequation is the set of all points on real line satisfying the given inequation. Therefore, the solution set of a system of linear inequations in one variable is the intersection of the solution sets of the linear inequations in the given system.

We use the following algorithm to solve a system of linear inequations in one variable.

#### ALGORITHM

- STEP I** Obtain the system of linear inequations.
- STEP II** Solve each inequation and obtain their solution sets. Also, represent them on real time.
- STEP III** Find the intersection of the solution sets obtained in step II by taking the help of the graphical representation of the solution sets in step II.
- STEP IV** The set obtained in step III is the required solution set of the given system of inequations.

Following examples will illustrate the above algorithm.

#### ILLUSTRATIVE EXAMPLES

##### LEVEL-1

**EXAMPLE 1** Solve the following system of linear inequations:

$$3x - 6 \geq 0$$

$$4x - 10 \leq 6$$

**SOLUTION** The given system of inequations is

$$3x - 6 \geq 0$$

...(i)

$$4x - 10 \leq 6$$

...(ii)

Now,  $3x - 6 \geq 0 \Rightarrow 3x \geq 6 \Rightarrow \frac{3x}{3} \geq \frac{6}{3} \Rightarrow x \geq 2$

$\therefore$  Solution set of inequation (i) is  $[2, \infty)$

and,  $4x - 10 \leq 6 \Rightarrow 4x \leq 16 \Rightarrow x \leq 4$

$\therefore$  Solution set of inequation (ii) is  $(-\infty, 4]$

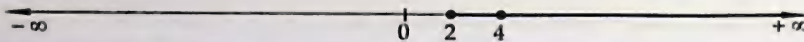


Fig. 15.16(i)

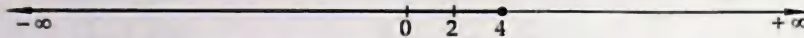


Fig. 15.16(ii)

The solution sets of inequations (i) and (ii) are represented graphically on real line in Figs. 15.16 (i) and (ii) respectively.

Clearly, the intersection of these solution sets is the set  $[2, 4]$ .

Hence, the solution set of the given system of inequations is the interval  $[2, 4]$ .

**EXAMPLE 2** Solve the following system of inequations:

$$\begin{aligned} \frac{5x}{4} + \frac{3x}{8} &> \frac{39}{8} \\ \frac{2x-1}{12} - \frac{x-1}{3} &< \frac{3x+1}{4} \end{aligned}$$

**SOLUTION** The given system of inequation is

$$\frac{5x}{4} + \frac{3x}{8} > \frac{39}{8} \quad \dots(i)$$

$$\frac{2x-1}{12} - \frac{x-1}{3} < \frac{3x+1}{4} \quad \dots(ii)$$

Now,

$$\Rightarrow \frac{5x}{4} + \frac{3x}{8} > \frac{39}{8}$$

$$\Rightarrow \frac{10x + 3x}{8} > \frac{39}{8}$$

$$\Rightarrow 13x > 39$$

$$\Rightarrow x > 3$$

$$\Rightarrow x \in (3, \infty)$$

So, the solution set of inequation (i) is the interval  $(3, \infty)$ .

and,

$$\frac{2x-1}{12} - \frac{x-1}{3} < \frac{3x+1}{4}$$

$$\Rightarrow \frac{(2x-1) - 4(x-1)}{12} < \frac{3x+1}{4}$$

$$\Rightarrow \frac{-2x+3}{12} < \frac{3x+1}{4}$$

$$\Rightarrow -2x+3 < 3(3x+1) \quad [\text{Multiplying both sides by 12 i.e. the l.c.m. of 12 and 4}]$$

$$\Rightarrow -2x+3 < 9x+3$$

$$\Rightarrow -2x-9x < 3-3$$

$$\Rightarrow -11x < 0$$

$$\Rightarrow x > 0$$

$$\Rightarrow x \in (0, \infty)$$

So, the solution set of inequation (ii) is the interval  $(0, \infty)$ . Let us now represent the solution sets of inequations (i) and (ii) on real line. These solution sets are graphed on real line in Figs. 15.17 (i) and 15.17 (ii) respectively.

From Figs. 15.17 (i) and (ii), we observe that the intersection of the solution sets of inequations (i) and (ii) is interval  $(3, \infty)$  represented by common thick line.

Hence, the solution set of the given system of inequations is the interval  $(3, \infty)$ .



Fig. 15.17(i)



Fig. 15.17(ii)

**EXAMPLE 3** Solve the following system of inequations:  $2(2x + 3) - 10 < 6(x - 2)$

$$\frac{2x - 3}{4} + 6 \geq 2 + \frac{4x}{3}$$

**SOLUTION** The given system of inequations is

$$2(2x + 3) - 10 < 6(x - 2) \quad \dots(i)$$

$$\frac{2x - 3}{4} + 6 \geq 2 + \frac{4x}{3} \quad \dots(ii)$$

Now,  $2(2x + 3) - 10 < 6(x - 2)$

$$\Rightarrow 4x + 6 - 10 < 6x - 12$$

$$\Rightarrow 4x - 4 < 6x - 12$$

$$\Rightarrow 4x - 6x < 4 - 12$$

$$\Rightarrow -2x < -8$$

$$\Rightarrow x > 4$$

$$\Rightarrow x \in (4, \infty)$$

So, the solution set of the first inequation is the interval  $(4, \infty)$ .

and,  $\frac{2x - 3}{4} + 6 \geq 2 + \frac{4x}{3}$

$$\Rightarrow \frac{2x - 3 + 24}{4} \geq \frac{6 + 4x}{3}$$

$$\Rightarrow \frac{2x + 21}{4} \geq \frac{4x + 6}{3}$$

$$\Rightarrow 3(2x + 21) \geq 4(4x + 6)$$

$$\Rightarrow 6x + 63 \geq 16x + 24$$

$$\Rightarrow 6x - 16x \geq 24 - 63$$

$$\Rightarrow -10x \geq -39$$

$$\Rightarrow x \leq \frac{39}{10}$$

$$\Rightarrow x \leq 3.9$$

$$\Rightarrow x \in (-\infty, 3.9]$$

So, the solution set of inequation (ii) is the interval  $(-\infty, 3.9]$ .

The solution sets of inequations (i) and (ii) are graphed on real line in Figs. 15.18 (i) and (ii) respectively.

We observe that there is no common solution of the two inequations. So, the given system of inequations has no solution.

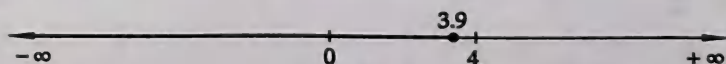


Fig. 15.18 (i)

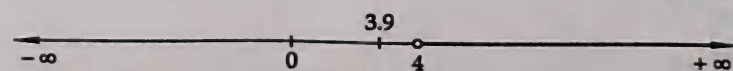


Fig. 15.18 (ii)

**EXAMPLE 4** Solve:  $-11 \leq 4x - 3 \leq 13$

**SOLUTION** We have,

$$-11 \geq 4x - 3 \geq 13 \Leftrightarrow -11 \geq 4x - 3 \text{ and } 4x - 3 \geq 13$$

Thus, we have two inequations and we wish to solve them simultaneously. Instead of solving these inequations by using the method discussed in first three examples, let us solve them directly in a different way as given below.

We have,

$$\begin{aligned} & -11 \leq 4x - 3 \leq 13 \\ \Rightarrow & -11 + 3 \leq 4x - 3 + 3 \leq 13 + 3 && \text{[Adding 3 throughout]} \\ \Rightarrow & -8 \leq 4x \leq 16 \\ \Rightarrow & \frac{-8}{4} \leq x \leq \frac{16}{4} && \text{[Dividing by 4 throughout]} \\ \Rightarrow & -2 \leq x \leq 4 \\ \Rightarrow & x \in [-2, 4] \end{aligned}$$

Hence, the interval  $[-2, 4]$  is the solution set of the given system of inequations.

**EXAMPLE 5** Solve:  $-5 \leq \frac{2-3x}{4} \leq 9$  **[NCERT EXEMPLAR]**

**SOLUTION** We have,

$$\begin{aligned} & -5 \leq \frac{2-3x}{4} \leq 9 \\ \Rightarrow & -5 \times 4 \leq \frac{2-3x}{4} \times 4 \leq 9 \times 4 && \text{[Multiplying throughout by 4]} \\ \Rightarrow & -20 \leq 2-3x \leq 36 \\ \Rightarrow & -20-2 \leq 2-3x-2 \leq 36-2 && \text{[Subtracting 2 throughout]} \\ \Rightarrow & -22 \leq -3x \leq 34 \\ \Rightarrow & \frac{-22}{-3} \geq \frac{-3x}{-3} \geq \frac{34}{-3} && \text{[Dividing throughout by -3]} \\ \Rightarrow & \frac{22}{3} \geq x \geq \frac{-34}{3} \\ \Rightarrow & \frac{-34}{3} \leq x \leq \frac{22}{3} \\ \Rightarrow & x \in [-34/3, 22/3] \end{aligned}$$

Hence, the interval  $[-34/3, 22/3]$  is the solution set of the given system of inequations.

**EXAMPLE 6** Solve the system of inequations:  $\frac{x}{2x+1} \geq \frac{1}{4}$ ,  $\frac{6x}{4x-1} < \frac{1}{2}$

**SOLUTION** The given system of inequations is

$$\frac{x}{2x+1} \geq \frac{1}{4} \quad \dots(i)$$

$$\frac{6x}{4x-1} < \frac{1}{2} \quad \dots(ii)$$

Now,  $\frac{x}{2x+1} \geq \frac{1}{4}$

$$\Rightarrow \frac{x}{2x+1} - \frac{1}{4} \geq 0$$

$$\Rightarrow \frac{4x - (2x+1)}{4(2x+1)} \geq 0$$

$$\Rightarrow \frac{2x-1}{2x+1} \geq 0 \quad [\text{Multiplying both sides by 4}]$$

$$\Rightarrow x \in (-\infty, -1/2) \cup [1/2, \infty) \quad [\text{See Fig. 15.19 (i)}]$$

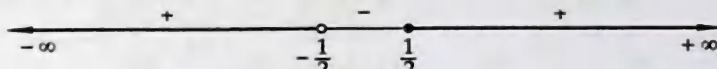


Fig. 15.19(i)

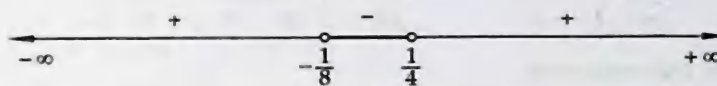


Fig. 15.19(ii)

Thus, the solution set of inequation (i) is  $(-\infty, -1/2) \cup [1/2, \infty)$  ... (iii)

And,  $\frac{6x}{4x-1} < \frac{1}{2}$

$$\Rightarrow \frac{6x}{4x-1} - \frac{1}{2} < 0$$

$$\Rightarrow \frac{12x - (4x-1)}{2(4x-1)} < 0$$

$$\Rightarrow \frac{8x+1}{2(4x-1)} < 0$$

$$\Rightarrow \frac{8x+1}{4x-1} < 0$$

[Multiplying both sides by 2]

$$\Rightarrow x \in (-1/8, 1/4) \quad [\text{See Fig. 15.19 (ii)}]$$

Thus, the solution set of inequation (ii) is  $(-1/8, 1/4)$  ... (iv)

It is evident from Fig. 15.19 that the intersection of (iii) and (iv) is the null set.

Hence, the given system of equations has no solution.

### EXERCISE 15.2

#### LEVEL-1

Solve each of the following system of equations in  $R$ .

1.  $x + 3 > 0$ ,  $2x < 14$
2.  $2x - 7 > 5 - x$ ,  $11 - 5x \leq 1$
3.  $x - 2 > 0$ ,  $3x < 18$
4.  $2x + 6 \geq 0$ ,  $4x - 7 < 0$
5.  $3x - 6 > 0$ ,  $2x - 5 > 0$
6.  $2x - 3 < 7$ ,  $2x > -4$
7.  $2x + 5 \leq 0$ ,  $x - 3 \leq 0$
8.  $5x - 1 < 24$ ,  $5x + 1 > -24$
9.  $3x - 1 \geq 5$ ,  $x + 2 > -1$
10.  $11 - 5x > -4$ ,  $4x + 13 \leq -11$
11.  $4x - 1 \leq 0$ ,  $3 - 4x < 0$
12.  $x + 5 > 2(x + 1)$ ,  $2 - x < 3(x + 2)$
13.  $2(x - 6) < 3x - 7$ ,  $11 - 2x < 6 - x$
14.  $5x - 7 < 3(x + 3)$ ,  $1 - \frac{3x}{2} \geq x - 4$
15.  $\frac{2x-3}{4} - 2 \geq \frac{4x}{3} - 6$ ,  $2(2x+3) < 6(x-2) + 10$
16.  $\frac{7x-1}{2} < -3$ ,  $\frac{3x+8}{5} + 11 < 0$
17.  $\frac{2x+1}{7x-1} > 5$ ,  $\frac{x+7}{x-8} > 2$
18.  $0 < \frac{-x}{2} < 3$
19.  $10 \leq -5(x-2) < 20$
20.  $-5 < 2x - 3 < 5$

$$21. \frac{4}{x+1} \leq 3 \leq \frac{6}{x+1}, x > 0$$

[NCERT EXEMPLAR]

**ANSWERS**

- |                    |                     |                      |                      |
|--------------------|---------------------|----------------------|----------------------|
| 1. $(-3, 7)$       | 2. $(4, \infty)$    | 3. $(2, 6)$          | 4. $[-3, 7/4)$       |
| 5. $(5/2, \infty)$ | 6. $(-2, 5)$        | 7. $(-\infty, -5/2]$ | 8. $(-5, 5)$         |
| 9. $[2, \infty)$   | 10. $(-\infty, -6]$ | 11. No Solution      | 12. $(-1, 3)$        |
| 13. $(5, \infty)$  | 14. $(-\infty, 2]$  | 15. No Solution      | 16. $(-\infty, -21)$ |
| 17. No Solution    | 18. $(-6, 0)$       | 19. $(-2, 0]$        | 20. $(-1, 4)$        |
|                    |                     |                      | 21. $[1/3, 1]$       |

**15.5.1 SOME IMPORTANT RESULTS**

In this sub-section, let us discuss some results on inequations involving modulus of the variable. We state and prove these results as theorems.

**THEOREM 1** If  $a$  is a positive real number, then

- (i)  $|x| < a \Leftrightarrow -a < x < a$  i.e.  $x \in (-a, a)$   
 (ii)  $|x| \leq a \Leftrightarrow -a \leq x \leq a$  i.e.  $x \in [-a, a]$



Fig. 15.20(i)

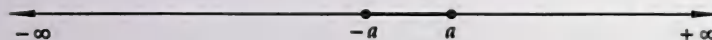


Fig. 15.20(ii)

**PROOF (i)** We know that:  $|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$

So, we consider the following cases:

**CASE I** When  $x \geq 0$ : In this case,  $|x| = x$ .

$$\therefore |x| < a \Rightarrow x < a$$

Thus, in this case the solution set of the given inequation is given by

$$x \geq 0 \text{ and } x < a \Rightarrow 0 \leq x < a \quad \dots(i)$$

**CASE II** When  $x < 0$ : In this case,  $|x| = -x$ .

$$\therefore |x| < a \Rightarrow -x < a \Rightarrow x > -a$$

Thus, in this case the solution set of the given inequation is given by

$$x < 0 \text{ and } x > -a \Rightarrow -a < x < 0 \quad \dots(ii)$$

Combining (i) and (ii), we get

$$|x| < a \Leftrightarrow -a < x < 0 \text{ or } 0 \leq x < a \Leftrightarrow -a < x < a.$$

(ii) Proceeding exactly as in (i), we get

$$|x| \leq a \Rightarrow -a \leq x \leq a.$$

**THEOREM 2** If  $a$  is a positive real number, then

- (i)  $|x| > a \Leftrightarrow x < -a \text{ or } x > a$

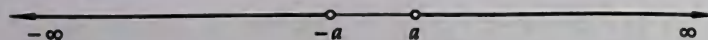


Fig. 15.21(i)

- (ii)  $|x| \geq a \Leftrightarrow x \leq -a \text{ or } x \geq a$



Fig. 15.21(ii)

**PROOF CASE I** When  $x > 0$ : In this case,  $|x| = x$

$$\therefore |x| > a \Rightarrow x > a$$

Thus, in this case the solution set of the inequality  $|x| > a$  is given by

$$x > 0 \text{ and } x > a \Rightarrow x > a$$

$$[\because a > 0] \dots(i)$$

**CASE II** When  $x < 0$ : In this case,  $|x| = -x$

$$\therefore |x| > a \Rightarrow -x > a \Rightarrow x < -a$$

Thus, in this case the solution set of the given inequality is given by

$$x < 0 \text{ and } x < -a \Rightarrow x < -a$$

$$[\because a > 0] \dots(ii)$$

Combining (i) and (ii), we get

$$|x| > a \Leftrightarrow x < -a \text{ or } x > a$$

(ii) Proceeding as in (i), we get

$$|x| \geq a \Leftrightarrow x \leq -a \text{ or } x \geq a.$$

**THEOREM 3** Let  $r$  be a positive real number and  $a$  be a fixed real number. Then,

$$(i) |x - a| < r \Leftrightarrow a - r < x < a + r \text{ i.e. } x \in (a - r, a + r)$$

$$(ii) |x - a| \leq r \Leftrightarrow a - r \leq x \leq a + r \text{ i.e. } x \in [a - r, a + r]$$

$$(iii) |x - a| > r \Leftrightarrow x < a - r, \text{ or } x > a + r$$

$$(iv) |x - a| \geq r \Leftrightarrow x \leq a - r, \text{ or } x \geq a + r$$

**PROOF** (i) Using Theorem 1, we obtain

$$|x - a| < r \Leftrightarrow -r < x - a < r \Leftrightarrow a - r < x - a + a < a + r \Leftrightarrow a - r < x < a + r$$

(ii) Using Theorem 1 (ii), we obtain

$$|x - a| \leq r \Leftrightarrow -r \leq x - a \leq r \Leftrightarrow a - r \leq x - a + a \leq a + r \Leftrightarrow a - r \leq x \leq a + r$$

(iii) Using Theorem 2(i), we obtain

$$|x - a| > r \Leftrightarrow x - a < -r, \text{ or } x - a > r \Leftrightarrow x < a - r, \text{ or } x > a + r$$

(iv) Using Theorem 2 (ii), we obtain,

$$|x - a| \geq r \Leftrightarrow x - a \leq -r, \text{ or } x - a \geq r \Leftrightarrow x \leq a - r, \text{ or } x \geq a + r$$

**NOTE:** These results may be used directly for solving linear inequalities involving absolute values.

**THEOREM 4** Let  $a, b$  be positive real numbers. Then

$$(i) a < |x| < b \Leftrightarrow x \in (-b, -a) \cup (a, b)$$

$$(ii) a \leq |x| \leq b \Leftrightarrow x \in [-b, -a] \cup [a, b]$$

$$(iii) a \leq |x - c| \leq b \Leftrightarrow x \in [-b + c, -a + c] \cup [a + c, b + c]$$

$$(iv) a < |x - c| < b \Leftrightarrow x \in (-b + c, -a + c) \cup (a + c, b + c)$$

**PROOF** (i)  $a < |x| < b \Leftrightarrow |x| > a \text{ and } |x| < b \Leftrightarrow (x < -a \text{ or } x > a) \text{ and } (-b < x < b)$

$$\Leftrightarrow x \in (-b, -a) \cup (a, b)$$

Similarly, we can prove other results.

### ILLUSTRATIVE EXAMPLES

#### LEVEL-2

**EXAMPLE 1**  $|3x - 2| \leq \frac{1}{2}$

**SOLUTION** We know that:  $|x - a| \leq r \Leftrightarrow a - r \leq x \leq a + r$

$$\therefore |3x - 2| \leq \frac{1}{2} \Leftrightarrow 2 - \frac{1}{2} \leq 3x \leq 2 + \frac{1}{2} \Leftrightarrow \frac{3}{2} \leq 3x \leq \frac{5}{2} \Leftrightarrow \frac{1}{2} \leq x \leq \frac{5}{6} \Leftrightarrow x \in [1/2, 5/6]$$

Hence, the solution set of the given inequality is the interval  $[1/2, 5/6]$ .

**EXAMPLE 2** Solve:  $|x - 2| \geq 5$

**SOLUTION** We know that:  $|x - a| \geq r \Leftrightarrow x \leq a - r, \text{ or } x \geq a + r$

$$\therefore |x - 2| \geq 5$$

$$\Leftrightarrow x \leq -5, \text{ or } x \geq 2 + 5$$

$$\Leftrightarrow x \leq -3 \text{ or } x \geq 7 \Leftrightarrow x \in (-\infty, -3] \text{ or } x \in [7, \infty) \Leftrightarrow x \in (-\infty, -3] \cup [7, \infty)$$

Hence the solution set of the given inequation is  $(-\infty, -3] \cup [7, \infty)$

**EXAMPLE 3** Solve :  $1 \leq |x - 2| \leq 3$

**SOLUTION** We know that :

$$a \leq |x - c| \leq b \Leftrightarrow x \in [-b + c, -a + c] \cup [a + c, b + c]$$

$$\therefore 1 \leq |x - 2| \leq 3 \Leftrightarrow x \in [-3 + 2, -1 + 2] \cup [1 + 2, 3 + 2] \Leftrightarrow x \in [-1, 1] \cup [3, 5]$$

Hence, the solution set of the given inequation is  $[-1, 1] \cup [3, 5]$ .

**EXAMPLE 4** Solve the following system of inequations:  $|x - 1| \leq 5, |x| \geq 2$  [NCERT EXEMPLAR]

**SOLUTION** The given system of inequations is

$$|x - 1| \leq 5 \quad \dots(i)$$

$$|x| \geq 2 \quad \dots(ii)$$

Now,  $|x - 1| \leq 5$

$$\Rightarrow 1 - 5 \leq x \leq 1 + 5$$

$$[\because |x - a| \leq r \Leftrightarrow a - r \leq x \leq a + r]$$

$$\Rightarrow -4 \leq x \leq 6 \Rightarrow x \in [-4, 6]$$

Thus, the solution set of (i) is the interval  $x \in [-4, 6]$ .

$$\text{and, } |x| \geq 2 \Leftrightarrow x \leq -2, \text{ or } x \geq 2$$

$$[\because |x| \geq a \Rightarrow x \leq -a \text{ or } x \geq a]$$

$$\Leftrightarrow x \in (-\infty, -2] \cup [2, \infty)$$

Thus, the solution set of (ii) is  $(-\infty, -2] \cup [2, \infty)$ .

The solution sets of inequations (i) and (ii) are represented graphically in Figures 15.22 (i) and 15.22 (ii) respectively. The intersection of these two is  $[-4, -2] \cup [2, 6]$

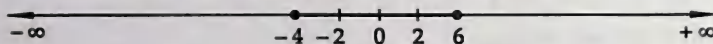


Fig. 15.22(i)

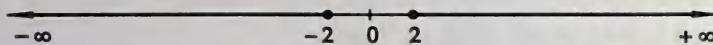


Fig. 15.22(ii)

Hence, the solution set of the given system of inequations is  $[-4, -2] \cup [2, 6]$ .

**EXAMPLE 5** Solve :  $\frac{|x| - 1}{|x| - 2} \geq 0, x \in R, x \neq \pm 2$ .

**SOLUTION** We have,

$$\frac{|x| - 1}{|x| - 2} \geq 0$$



Fig. 15.23

$$\Rightarrow \frac{y - 1}{y - 2} \geq 0, \text{ where } y = |x|$$

$$\Rightarrow y \leq 1 \text{ or } y > 2$$

[ See Fig. 15.23]

$$\Rightarrow |x| \leq 1 \text{ or } |x| > 2$$

$$\Rightarrow (-1 \leq x \leq 1) \text{ or } (x < -2 \text{ or } x > 2)$$

$$\Rightarrow x \in [-1, 1] \text{ or } x \in (-\infty, -2) \cup (2, \infty)$$

$$\Rightarrow x \in [-1, 1] \cup (-\infty, -2) \cup (2, \infty)$$

Hence, the solution set of the given inequation is  $[-1, 1] \cup (-\infty, -2) \cup (2, \infty)$ .

**EXAMPLE 6** Solve:  $\frac{-1}{|x| - 2} \geq 1, \text{ where } x \in R, x \neq \pm 2$

SOLUTION We have,  $\frac{-1}{|x|-2} \geq 1$

$$\Rightarrow \frac{-1}{|x|-2} - 1 \geq 0$$

$$\Rightarrow \frac{-1 - (|x|-2)}{|x|-2} \geq 0$$

$$\Rightarrow \frac{1-|x|}{|x|-2} \geq 0$$

$$\Rightarrow \frac{|x|-1}{|x|-2} \leq 0$$

$$\Rightarrow \frac{y-1}{y-2} \leq 0, \text{ where } y = |x|$$

$$\Rightarrow 1 \leq y < 2$$

$$\Rightarrow 1 \leq |x| < 2$$

$$\Rightarrow x \in (-2, -1] \cup [1, 2)$$

$$[\because a < |x| \leq b \Leftrightarrow x \in [-b, -a) \cup (a, b]]$$

[See Fig. 15.24]

$$[\because y = |x|]$$

Hence, the solution set of the given inequation is  $(-2, -1] \cup [1, 2)$



Fig. 15.24

EXAMPLE 7 Solve the inequation:  $\left| \frac{2}{x-4} \right| > 1, x \neq 4$ .

SOLUTION We have,

$$\left| \frac{2}{x-4} \right| > 1, x \neq 4$$

$$\Rightarrow \frac{2}{|x-4|} > 1$$

$$\left[ \because \left| \frac{a}{b} \right| = \frac{|a|}{|b|} \right]$$

$$\Rightarrow 2 > |x-4|$$

$$[\because |x-4| > 0 \text{ for all } x \neq 4]$$

$$\Rightarrow 4-2 < x < 4+2$$

$$[\because |x-a| < r \Leftrightarrow a-r < x < a+r]$$

$$\Rightarrow 2 < x < 6$$

$$\Rightarrow x \in (2, 6)$$

But,  $x \neq 4$ .

Hence, the solution set of the given inequation is  $(2, 4) \cup (4, 6)$ .

EXAMPLE 8 Solve:  $\frac{|x+3|+x}{x+2} > 1$

[NCERT EXEMPLAR]

SOLUTION We have,  $\frac{|x+3|+x}{x+2} > 1$ .

Clearly, LHS of this inequation is meaningful for  $x \neq -2$ .

Now,  $\frac{|x+3|+x}{x+2} > 1$

$$\Rightarrow \frac{|x+3|+x}{x+2} - 1 > 0$$

$$\Rightarrow \frac{|x+3|+x-x-2}{x+2} > 0$$

$$\Rightarrow \frac{|x+3|-2}{x+2} > 0.$$

Now two cases arise:

**CASE I** When  $x + 3 \geq 0$  i.e.  $x \geq -3$ : In this case,  $|x + 3| = x + 3$ .

$$\therefore \frac{|x + 3| - 2}{x + 2} > 0$$

$$\Rightarrow \frac{x + 3 - 2}{x + 2} > 0$$

$$\Rightarrow \frac{x + 1}{x + 2} > 0$$

$$\Rightarrow x \in (-\infty, -2) \cup (-1, \infty)$$

[ See Fig. 15.25]

But,  $x \geq -3$ . Therefore, the solution set of the given inequation in this case is  $[-3, -2) \cup (-1, \infty)$ .

**CASE II** When  $x + 3 < 0$  i.e.  $x < -3$ : In this case,  $|x + 3| = -(x + 3)$ .

$$\therefore \frac{|x + 3| - 2}{x + 2} > 0$$

$$\Rightarrow \frac{-(x + 3) - 2}{x + 2} > 0$$

$$\Rightarrow \frac{-(x + 5)}{x + 2} > 0$$

$$\Rightarrow \frac{x + 5}{x + 2} < 0$$

$$\Rightarrow x \in (-5, -2)$$

[See Fig. 15.26]

But,  $x < -3$ . Therefore, the solution set of the given inequation in this case is the interval  $(-5, -3)$ .

From Case I and Case II, we obtain that the solution set of the given inequation is

$$[-3, -2) \cup (-1, \infty) \cup (-5, -3) = (-5, -2) \cup (-1, \infty).$$

**EXAMPLE 9** Solve:  $|x - 1| + |x - 2| \geq 4$

**SOLUTION** On the LHS of the given inequation there are two terms both containing modulus. By equating the expressions within the modulus to zero, we get  $x = 1, 2$  as critical points. These points divide real line in three parts viz.  $(-\infty, 1]$ ,  $[1, 2]$  and  $[2, \infty)$ . So, we consider the following three cases.

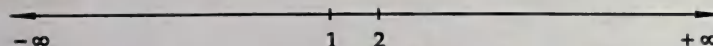


Fig. 15.27

**CASE I** When  $-\infty < x < 1$ : In this case, we have  $|x - 1| = -(x - 1)$  and  $|x - 2| = -(x - 2)$

$$\therefore |x - 1| + |x - 2| \geq 4$$

$$\Rightarrow -(x - 1) - (x - 2) \geq 4$$

$$\Rightarrow -2x + 3 \geq 4$$

$$\Rightarrow -2x \geq 1$$

$$\Rightarrow x \leq -\frac{1}{2}$$

But,  $-\infty < x < 1$ . Therefore, in this case the solution set of the given inequation is  $(-\infty, -1/2]$

**CASE II** When  $1 \leq x < 2$ : In this case, we have  $|x - 1| = (x - 1)$  and  $|x - 2| = -(x - 2)$

$$\therefore |x - 1| + |x - 2| \geq 4$$

$$\Rightarrow x - 1 - (x - 2) \geq 4$$

$$\Rightarrow 1 \geq 4, \text{ which is an absurd result.}$$

So, the given inequation has no solution for  $x \in [1, 2)$ .

**CASE III** When  $x \geq 2$ : In this case, we have  $|x - 1| = x - 1$  and  $|x - 2| = x - 2$

$$\therefore |x - 1| + |x - 2| \geq 4$$

$$\Rightarrow x - 1 + x - 2 \geq 4$$

$$\Rightarrow 2x - 3 \geq 4$$

$$\Rightarrow 2x \geq 7$$

$$\Rightarrow x \geq \frac{7}{2}$$

But,  $x > 2$ . Therefore, in this case the solution set of the given inequation is  $[7/2, \infty)$ .

Combining Case I and Case II, we obtain that the solution set of the given inequation is  $(-\infty, -1/2] \cup [7/2, \infty)$

**EXAMPLE 10** Solve:  $\frac{|x - 1|}{x + 2} < 1$ .

**SOLUTION** We have,

$$\frac{|x - 1|}{x + 2} < 1 \Rightarrow \frac{|x - 1|}{x + 2} - 1 < 0 \Rightarrow \frac{|x - 1| - (x + 2)}{x + 2} < 0$$

Now the following cases arise.

**CASE I** When  $x - 1 \geq 0$  i.e.  $x \geq 1$ : In this case, we have  $|x - 1| = x - 1$

$$\therefore \frac{|x - 1| - (x + 2)}{x + 2} < 0$$

$$\Rightarrow \frac{(x - 1) - (x + 2)}{x + 2} < 0$$

$$\Rightarrow \frac{-3}{x + 2} < 0$$

$$\Rightarrow x + 2 > 0 \quad \left[ \because \frac{a}{b} < 0 \text{ and } a < 0 \Rightarrow b > 0 \right]$$

$$\Rightarrow x > -2$$

But,  $x \geq 1$ . Therefore,  $x > -2$  and  $x \geq 1$  implies that  $x \geq 1$ . Thus, in this case the solution set of the given inequation is  $[1, \infty)$ .

**CASE II** When  $x - 1 < 0$  i.e.  $x < 1$ : In this case, we have  $|x - 1| = -(x - 1)$ .

$$\therefore \frac{|x - 1| - (x + 2)}{x + 2} < 0$$

$$\Rightarrow \frac{-(x - 1) - (x + 2)}{x + 2} < 0$$

$$\Rightarrow \frac{-2x + 1}{x + 2} < 0$$

$$\Rightarrow \frac{2x + 1}{x + 2} > 0$$

$$\Rightarrow x \in (-\infty, -2) \cup (-1/2, \infty)$$

[See Fig. 15.28]

But,  $x < 1$ . Therefore,  $x \in (-\infty, -2) \cup (-1/2, \infty)$  and  $x < 1$  implies that  $x \in (-\infty, -2) \cup \left(-\frac{1}{2}, 1\right)$ .

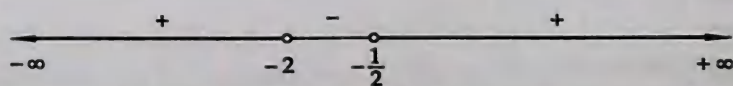


Fig. 15.28

Thus, in this case the solution set of the given inequation is  $(-\infty, -2) \cup (-1/2, 1)$ .

Combining Case I and Case II, we obtain that the solution set of the given inequation is  $(-\infty, -2) \cup (-1/2, \infty)$

**EXERCISE 15.3****LEVEL-2**

Solve each of the following system of equations in  $R$ .

1.  $\left|x + \frac{1}{3}\right| > \frac{8}{3}$

2.  $|4 - x| + 1 < 3$

3.  $\left|\frac{3x-4}{2}\right| \leq \frac{5}{12}$

4.  $\frac{|x-2|}{x-2} > 0$

5.  $\frac{1}{|x|-3} < \frac{1}{2}$

6.  $\frac{|x+2|-x}{x} < 2$

7.  $\left|\frac{2x-1}{x-1}\right| > 2$

8.  $|x-1| + |x-2| + |x-3| \geq 6$

9.  $\frac{|x-2|-1}{|x-2|-2} \leq 0$  [NCERT EXEMPLAR]

10.  $\frac{1}{|x|-3} \leq \frac{1}{2}$  [NCERT EXEMPLAR]

11.  $|x+1| + |x| > 3$  [NCERT EXEMPLAR]

12.  $1 \leq |x-2| \leq 3$

13.  $|3-4x| \geq 9$  [NCERT EXEMPLAR]

**ANSWERS**

1.  $(-\infty, -3) \cup (7/3, \infty)$

2.  $(2, 6)$

3.  $[19/18, 29/18]$

4.  $(2, \infty)$

5.  $(-\infty, -5) \cup (-3, 3) \cup (5, \infty)$

6.  $(-\infty, 0) \cup (1, \infty)$

7.  $(3/4, 1) \cup (1, \infty)$

8.  $(-\infty, 0] \cup [4, \infty)$

9.  $(0, 1] \cup [3, 4)$

10.  $(-\infty, -5] \cup (-3, 3) \cup [5, \infty)$

11.  $(-\infty, -2) \cup (1, \infty)$

12.  $[-1, 1] \cup [3, 5]$

13.  $(-\infty, -3/2] \cup [3, \infty)$

**15.6 SOME APPLICATIONS OF LINEAR IN EQUATIONS IN ONE VARIABLE**

In this section, we shall utilize the knowledge of solving linear in equations in one variable in solving different problems from various fields such as science, engineering, economics etc.

Following examples will illustrate the same

**ILLUSTRATIVE EXAMPLES****LEVEL-1**

**EXAMPLE 1** Find all pairs of consecutive odd positive integers, both of which are smaller than 18, such that their sum is more than 20.

**SOLUTION** Let  $x$  be the smaller of the two consecutive odd positive integers. Then, the other odd integer is  $x + 2$ .

It is given that both the integers are smaller than 18 and their sum is more than 20. Therefore,

$$x + 2 < 18 \text{ and } x + (x + 2) > 20$$

$$\Rightarrow x < 16 \text{ and } 2x + 2 > 20$$

$$\Rightarrow x < 16 \text{ and } 2x > 18$$

$$\Rightarrow x < 16 \text{ and } x > 9 \Rightarrow 9 < x < 16 \Rightarrow x = 11, 13, 15$$

[ $\because x$  is an odd integer]

Hence, the required pairs of odd integers are  $(11, 13)$ ,  $(13, 15)$  and  $(15, 17)$ .

**EXAMPLE 2** Find all pairs of consecutive even positive integers, both of which are larger than 8, such that their sum is less than 25.

**SOLUTION** Let  $x$  be the smaller of the two consecutive even positive integers. Then, the other even integer is  $x + 2$ .

It is given that both the integers are larger than 8 and their sum is less than 25. Therefore,

$$x > 8 \text{ and } x + x + 2 < 25$$

$$\Rightarrow x > 8 \text{ and } 2x + 2 < 25$$

$$\Rightarrow x > 8 \text{ and } 2x < 23$$

$$\Rightarrow x > 8 \text{ and } x < \frac{23}{2} \Rightarrow 8 < x < \frac{23}{2} \Rightarrow x = 10 \quad [\because x \text{ is an even integer}]$$

Hence, the required pair of even integers is (10, 12).

**EXAMPLE 3** The cost and revenue functions of a product are given by  $C(x) = 2x + 400$  and  $R(x) = 6x + 20$  respectively, where  $x$  is the number of items produced by the manufacturer. How many items the manufacturer must sell to realize some profit?

**SOLUTION** We know that: Profit = Revenue - Cost. Therefore, to earn some profit, we must have

$$\text{Revenue} > \text{Cost}$$

$$\Rightarrow 6x + 20 > 2x + 400$$

$$\Rightarrow 6x - 2x > 400 - 20 \Rightarrow 4x > 380 \Rightarrow x > \frac{380}{4} = 95$$

Hence, the manufacturer must sell more than 95 items to realize some profit.

**EXAMPLE 4** IQ of a person is given by the formula:  $IQ = \frac{MA}{CA} \times 100$ , where MA is mental age and CA is chronological age. If  $80 \leq IQ \leq 140$  for a group of 12 year children, find the range of their mental age.

**SOLUTION** We have: CA = 12 years

$$\therefore IQ = \frac{MA}{CA} \times 100 \Rightarrow IQ = \frac{MA}{12} \times 100 = \frac{25}{3} MA$$

$$\text{Now, } 80 \leq IQ \leq 140$$

$$\Rightarrow 80 \leq \frac{25}{3} MA \leq 140$$

$$\Rightarrow 240 \leq 25 MA \leq 420 \Rightarrow \frac{240}{25} \leq MA \leq \frac{420}{25} \Rightarrow 9.6 \leq MA \leq 16.8$$

**EXAMPLE 5** In the first four papers each of 100 marks, Rishi got 95, 72, 73, 83 marks. If he wants an average of greater than or equal to 75 marks and less than 80 marks, find the range of marks he should score in the fifth paper.

**SOLUTION** Suppose scores  $x$  marks in the fifth paper. Then,

$$75 \leq \frac{95 + 72 + 73 + 83 + x}{5} < 80$$

$$\Rightarrow 75 \leq \frac{323 + x}{5} < 80 \Rightarrow 375 < 323 + x < 400 \Rightarrow 52 < x < 77$$

Hence, Rishi must score between 52 and 77 marks.

**EXAMPLE 6** A manufacturer has 600 litres of a 12% solution of acid. How many litres of a 30% acid solution must be added to it so that acid content in the resulting mixture will be more than 15% but less than 18%?

**SOLUTION** Let  $x$  litres of 30% acid solution be added to 600 litres of 12% solution of acid. Then,

Total quantity of mixture =  $(600 + x)$  litres

$$\text{Total acid content in the } (600 + x) \text{ litres of mixture} = \frac{30x}{100} + \frac{12}{100} \times 600$$

It is given that acid content in the resulting mixture must be more than 15% and less than 18%.

$$\therefore 15\% \text{ of } (600 + x) < \left( \frac{30x}{100} + \frac{12}{100} \times 600 \right) < 18\% \text{ of } (600 + x)$$

$$\Rightarrow \frac{15}{100} \times (600 + x) < \frac{30x}{100} + \frac{12}{100} \times 600 < \frac{18}{100} \times (600 + x)$$

$$\Rightarrow 15(600 + x) < 30x + 12 \times 600 < 18(600 + x) \quad [\text{Multiplying through out by } 100]$$

$$\Rightarrow 9000 + 15x < 30x + 7200 < 10800 + 18x$$

$$\Rightarrow 9000 + 15x < 30x + 7200 \text{ and } 30x + 7200 < 10800 + 18x$$

$$\Rightarrow 9000 - 7200 < 30x - 15x \text{ and } 30x - 18x < 10800 - 7200$$

$$\Rightarrow 1800 < 15x \text{ and } 12x < 3600$$

$$\Rightarrow 15x > 1800 \text{ and } 12x < 3600$$

$$\Rightarrow x > 120 \text{ and } x < 300$$

$$\Rightarrow 120 < x < 300$$

Hence, the number of litres of the 30% solution of acid must be more than 120 but less than 300.

**EXAMPLE 7** A man wants to cut three lengths from a single piece of board of length 91 cm. The second length is to be 3 cm longer than the shortest and third length is to be twice as long as the shortest. What are the possible lengths for the shortest board if third piece is to be at least 5 cm longer than the second?

**SOLUTION** Let the length of the shortest piece be  $x$  cm. Then, the lengths of the second and third piece are  $x + 3$  cm and  $2x$  cm respectively. Then,

$$x + (x + 3) + 2x \leq 91 \text{ and } 2x \geq (x + 3) + 5$$

$$\Rightarrow 4x + 3 \leq 91 \text{ and } 2x \geq x + 8$$

$$\Rightarrow 4x \leq 88 \text{ and } x \geq 8 \Rightarrow x \leq 22 \text{ and } x \geq 8 \Rightarrow 8 \leq x \leq 22.$$

Hence, the shortest piece must be at least 8 cm long but not more than 22 cm long.

### EXERCISE 15.4

#### LEVEL-1

- Find all pairs of consecutive odd positive integers, both of which are smaller than 10, such that their sum is more than 11.
- Find all pairs of consecutive odd natural number, both of which are larger than 10, such that their sum is less than 40.
- Find all pairs of consecutive even positive integers, both of which are larger than 5, such that their sum is less than 23.
- The marks scored by Rohit in two tests were 65 and 70. Find the minimum marks he should score in the third test to have an average of at least 65 marks.
- A solution is to be kept between  $86^\circ$  and  $95^\circ\text{F}$ . What is the range of temperature in degree Celsius, if the Celsius (C)/Fahrenheit (F) conversion formula is given by  $F = \frac{9}{5}C + 32$ .
- A solution is to be kept between  $30^\circ\text{C}$  and  $35^\circ\text{C}$ . What is the range of temperature in degree Fahrenheit?
- To receive grade 'A' in a course, one must obtain an average of 90 marks or more in five papers each of 100 marks. If Shikha scored 87, 95, 92 and 94 marks in first four papers, find the minimum marks that she must score in the last paper to get grade 'A' in the course.
- A company manufactures cassettes and its cost and revenue functions for a week are  $C = 300 + \frac{3}{2}x$  and  $R = 2x$  respectively, where  $x$  is the number of cassettes produced and sold in a week. How many cassettes must be sold for the company to realize a profit?

9. The longest side of a triangle is three times the shortest side and the third side is 2 cm shorter than the longest side if the perimeter of the triangles at least 61 cm, Find the minimum length of the shortest-side.
10. How many litres of water will have to be added to 1125 litres of the 45% solution of acid so that the resulting mixture will contain more than 25% but less than 30% acid content?
11. A solution of 8% boric acid is to be diluted by adding a 2% boric acid solution to it. The resulting mixture is to be more than 4% but less than 6% boric acid. If there are 640 litres of the 8% solution, how many litres of 2% solution will have to be added?
12. The water acidity in a pool is considered normal when the average pH reading of three daily measurements is between 7.2 and 7.8. If the first two pH reading are 7.48 and 7.85, find the range of pH value for the third reading that will result in the acidity level being normal.

**ANSWERS**

- |                                                     |                                           |                              |
|-----------------------------------------------------|-------------------------------------------|------------------------------|
| 1. (5,7), (7, 9)                                    | 2. (11, 13), (13, 15), (15, 17), (17, 19) | 3. (6, 8), (8, 10), (10, 12) |
| 4. 60                                               | 5. Between 30°C and 35°C                  | 6. Between 86°F and 95°F     |
| 7. 82 marks                                         | 8. More than 600                          | 9. 9 cm                      |
| 10. More than 562.5 litres but less than 900 litres |                                           |                              |
| 11. More than 320 litres but less than 1280 litres  |                                           | 12. Between 6.27 and 8.07    |

**15.7 GRAPHICAL SOLUTION OF LINEAR INEQUATIONS IN TWO VARIABLES**

If  $a, b, c$  are real numbers, then the equation  $ax + by + c = 0$  is called a linear equation in two variables  $x$  and  $y$  whereas the inequalities  $ax + by \leq c$ ,  $ax + by \geq c$ ,  $ax + by < c$  and  $ax + by > c$  are called linear inequations in two variables  $x$  and  $y$ .

We have studied in coordinate geometry that the graph of the equation  $ax + by = c$  is a straight line which divides the  $xy$ -plane into two parts which are represented by  $ax + by \leq c$  and  $ax + by \geq c$ . These two parts are known as the closed half-spaces. The regions represented by  $ax + by < c$  and  $ax + by > c$  are known as the open half spaces. In set theoretical notations, the set  $\{(x, y) : ax + by = c\}$  is the straight line, sets  $\{(x, y) : ax + by \leq c\}$  and  $\{(x, y) : ax + by \geq c\}$  are closed half spaces and the sets  $\{(x, y) : ax + by < c\}$  and  $\{(x, y) : ax + by > c\}$  are open half-spaces. These half spaces are also known as the solution sets of the corresponding inequations.

In order to find the solution set of a linear inequation in two variables, we follow the following algorithm.

**ALGORITHM**

- STEP I** Convert the given inequation, say  $ax + by \leq c$ , into the equation  $ax + by = c$  which represents a straight line in  $xy$ -plane.
- STEP II** Put  $y = 0$  in the equation obtained in step I to get the point where the line meets with  $x$ -axis. Similarly, put  $x = 0$  to obtain a point where the line meets with  $y$ -axis.
- STEP III** Join the points obtained in step II to obtain the graph of the line obtained from the given inequation. In case of a strict inequality i.e.  $ax + by < c$  or  $ax + by > c$ , draw the dotted line, otherwise mark it thick line.
- STEP IV** Choose a point, if possible  $(0, 0)$ , not lying on this line : Substitute its coordinates in the inequation. If the inequation is satisfied, then shade the portion of the plane which contains the chosen point; otherwise shade the portion which does not contain the chosen point.
- STEP V** The shaded region obtained in step IV represents the desired solution set.

**REMARK** In case of the inequalities  $ax + by \leq c$  and  $ax + by \geq c$  points on the line are also a part of the shaded region while in case of inequalities  $ax + by < c$  and  $ax + by > c$  points on the line  $ax + by = c$  are not in the shaded region.

The following examples illustrate the above algorithm.

## ILLUSTRATIVE EXAMPLES

## LEVEL-1

**EXAMPLE 1** Solve the following inequations graphically:

(i)  $2x + 3y \leq 6$

(ii)  $2x - y \geq 1$

(iii)  $x \geq 2$

(iv)  $y \leq -3$

**SOLUTION** (i) Converting the given inequation into equation, we obtain  $2x + 3y = 6$ .

Putting  $y = 0$  and  $x = 0$  respectively in this equation, we get  $x = 3$  and  $y = 2$ . So, this line meets  $x$ -axis at  $A(3, 0)$  and  $y$ -axis at  $B(0, 2)$ . We plot these points and join them by a thick line. This line divides the  $xy$ -plane in two parts. To determine the region represented by the given inequality consider the point  $O(0, 0)$ . Clearly,  $(0, 0)$  satisfies the inequality. So, the region containing the origin is represented by the given inequation as shown in Fig. 15.29. This region represents the solution set of the given inequations.

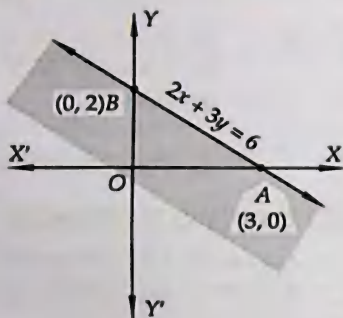


Fig. 15.29

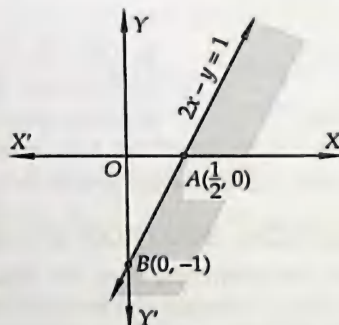


Fig. 15.30

(ii) Converting the given inequation into equation we obtain  $2x - y = 1$ . This line meets  $x$  and  $y$ -axes at  $A(1/2, 0)$  and  $B(0, -1)$  respectively. Joining these points by a thick line we obtain the line passing through  $A$  and  $B$  as shown in Fig. 15.30. This line divides the  $xy$ -plane into two regions viz. one lying above it and the other lying below it. Consider the point  $O(0, 0)$ . Clearly,  $(0, 0)$  does not satisfy the inequation  $2x - y \geq 1$ . So, the region not containing the origin is represented by the given inequation as shown in Fig. 15.30. Clearly it represents the solution set of the given inequation.

(iii) We have  $x \geq 2$ . Converting the inequation into equation, we obtain  $x = 2$ . Clearly, it is a line parallel to  $y$ -axis at a distance of 2 units from it. This line divides the  $xy$ -plane into two parts viz. one part on the LHS of  $x = 2$  and the other on its RHS. We find that the point  $(0, 0)$  does not satisfy the inequation  $x \geq 2$ . So, the region represented by the given equation is the shaded region shown in Fig. 15.31. The shaded region is the required solution set of the given inequation.

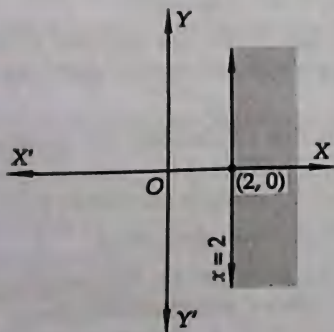


Fig. 15.31

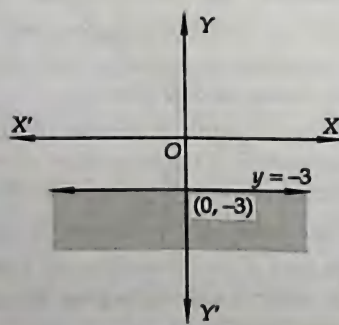


Fig. 15.32

(iv) We have  $y \leq -3$ . Converting the given inequation into equation we obtain  $y = -3$ . Clearly, it is a line parallel to  $x$ -axis at a distance of 3 units below it. The line  $y = -3$  divides the  $xy$ -plane into two regions one below it and the other above it. Consider the point  $O(0, 0)$ . We find that  $(0, 0)$  does not satisfy the inequation  $y \leq -3$ . So, the region represented by the given inequation is the region not containing the origin as shown in Fig. 15.32. Clearly, it is the solution set of the given inequation.

**EXAMPLE 2** Solve the following inequations graphically:

- (i)  $|x| \leq 3$       (ii)  $|y - x| \leq 3$       (iii)  $|x - y| \geq 1$

**SOLUTION** (i) Converting the given inequation into equation, we obtain  $x = 3$ . This equation represents a line parallel to  $y$ -axis at a distance of 3 units from it. The line given by  $x = 3$  divides the  $xy$ -plane into two regions. Clearly, the point  $O(0, 0)$  satisfies  $x \leq 3$ . So, the graph of  $x \leq 3$  is as shown in Fig. 15.33. The shaded region represents the solution set of this inequation.

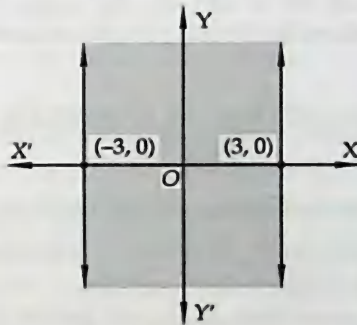


Fig. 15.33

(ii) We have,  $|y - x| \leq 3$ . This inequation is equivalent to

$$-3 \leq y - x \leq 3 \quad [\because |x| \leq a \Leftrightarrow -a \leq x \leq a]$$

$$\Leftrightarrow -3 \leq y - x \text{ and } y - x \leq 3$$

$$\Leftrightarrow x - y - 3 \leq 0 \text{ and } x - y + 3 \geq 0$$

The region represented by  $|y - x| \leq 3$  is the region common to the regions represented by  $x - y - 3 \leq 0$  and  $x - y + 3 \geq 0$  as shown in Fig. 15.34. This shaded region represents the solution set of the given inequation.

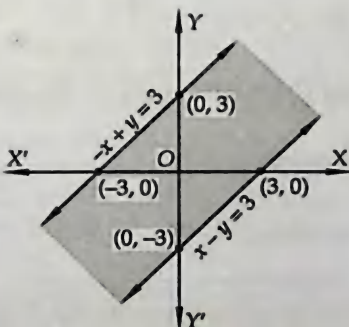


Fig. 15.34

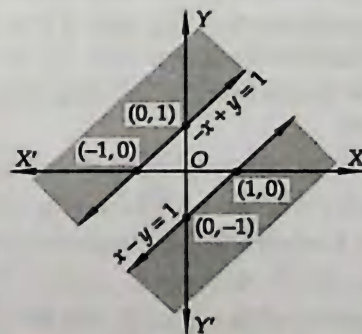


Fig. 15.35

(iii) We have,

$$|x - y| \geq 1 \Leftrightarrow x - y \geq 1 \text{ or } x - y \leq -1 \Leftrightarrow x - y - 1 \geq 0 \text{ or } x - y + 1 \leq 0$$

The required region is the union of regions represented by  $x - y - 1 \geq 0$  and  $x - y + 1 \leq 0$  as shown in Fig. 15.35. The shaded region represents the solution set of the given inequation.

## EXERCISE 15.5

## LEVEL-1

Represent to solution set of each of the following inequations graphically in two dimensional plane:

- |                              |                      |                    |
|------------------------------|----------------------|--------------------|
| 1. $x + 2y - y \leq 0$       | 2. $x + 2y \geq 6$   | 3. $x + 2 \geq 0$  |
| 4. $x - 2y < 0$              | 5. $-3x + 2y \leq 6$ | 6. $x \leq 8 - 4y$ |
| 7. $0 \leq 2x - 5y + 10$     | 8. $3y > 6 - 2x$     | 9. $y > 2x - 8$    |
| 10. $3x - 2y \leq x + y - 8$ |                      |                    |

## 15.8 SOLUTION OF SIMULTANEOUS LINEAR INEQUATIONS IN TWO VARIABLE

In this section, we will discuss the technique of finding the solution set of simultaneous linear inequations. Solving simultaneous linear inequations means finding the set of points  $(x, y)$  for which all the constraints are satisfied. Note that the solution set of simultaneous linear inequations may be an empty set or it may be the region bounded by the straight lines corresponding to linear inequations or it may be an unbounded region with straight line boundaries.

## ILLUSTRATIVE EXAMPLES

## LEVEL-1

**Type I ON FINDING THE SOLUTION SET REPRESENTED BY SIMULTANEOUS LINEAR INEQUATIONS**

**EXAMPLE 1** Exhibit graphically the solution set of the linear inequations

$$3x + 4y \leq 12, \quad 4x + 3y \leq 12, \quad x \geq 0, \quad y \geq 0$$

**SOLUTION** Converting the inequations into equations, the inequations reduce to

$$3x + 4y = 12, \quad 4x + 3y = 12, \quad x = 0 \text{ and } y = 0.$$

**Region Represented by  $3x + 4y \leq 12$ :** The line  $3x + 4y = 12$  meets the coordinate axes at A (4, 0) and B (0, 3). Draw a thick line joining A and B. We find that (0, 0) satisfies inequation  $3x + 4y \leq 12$ . So, the portion containing the origin represents the solution set of the inequation  $3x + 4y \leq 12$ .

**Region Represented by  $4x + 3y \leq 12$ :** The line  $4x + 3y = 12$  meets the x and y-axes at  $A_1(3, 0)$  and  $B_1(0, 4)$  respectively. Join these two points by a thick line. Clearly, the region containing the origin is represented by the inequation  $4x + 3y \leq 12$ .

**Region Represented by  $x \geq 0$  and  $y \geq 0$ :** Clearly,  $x \geq 0$  and  $y \geq 0$  represent the first quadrant.

Hence, the shaded region given in Fig. 15.36 represents the solution set of the given linear inequations.

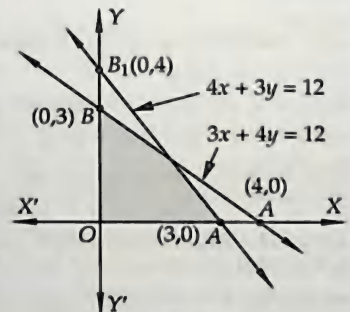


Fig. 15.36

**EXAMPLE 2** Exhibit graphically the solution set of the linear inequations

$$x + y \leq 5, \quad 4x + y \geq 4, \quad x + 5y \geq 5, \quad x \leq 4, \quad y \leq 3$$

**SOLUTION** Converting the inequations into equations, we obtain

$$x + y = 5, \quad 4x + y = 4, \quad x + 5y = 5, \quad x = 4, \quad y = 3$$

**Region Represented by  $x + y \leq 5$ :** The line  $x + y = 5$  meets the coordinate axes at A(5, 0) and B(0, 5) respectively. Join these points by a thick line. Clearly, (0, 0) satisfies the inequality  $x + y \leq 5$ . So, the portion containing the origin represents the solution set of the inequation  $x + y \leq 5$ .

**Region Represented by  $4x + y \geq 4$ :** The line  $4x + y = 4$  meets the coordinate axes at  $A_1(1, 0)$  and  $B_1(0, 4)$  respectively. Join these points by a thick line. Clearly, (0, 0) does not satisfy the inequation  $4x + y \geq 4$ . So, the portion not containing the origin is represented by the inequation  $4x + y \geq 4$ .

**Region Represented by  $x + 5y \geq 5$ :** The line  $x + 5y = 5$  meets the coordinate axes at  $A(5, 0)$  and  $B_2(0, 1)$  respectively. Join these two points by a thick line. We find that  $(0, 0)$  does not satisfy the inequation  $x + 5y \geq 5$ . So, the portion not containing the origin is represented by the given inequation.

**Region Represented by  $x \leq 4$ :** Clearly,  $x = 4$  is a line parallel to  $y$ -axis at a distance of 4 units from the origin. Since  $(0, 0)$  satisfies the inequation  $x \leq 4$ . So, the portion lying on the left side of  $x = 4$  is the region represented by  $x \leq 4$ .

**Region Represented by  $y \leq 3$ :** Clearly,  $y = 3$  is a line parallel to  $x$ -axis at a distance 3 from it. Since  $(0, 0)$  satisfies  $y \leq 3$ . So, the portion containing the origin is represented by the given inequation.

The common region of the above five regions represents the solution set of the given linear constraints as shown in Fig. 15.37.

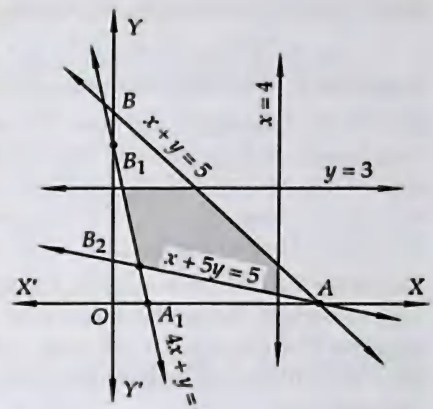


Fig. 15.37

**EXAMPLE 3** Draw the diagram of the solution set of the linear inequations  $3x + 4y \geq 12$ ,  $y \geq 1$ ,  $x \geq 0$ .

**SOLUTION** Converting the inequations into equations, we get  $3x + 4y = 12$ ,  $y = 1$ ,  $x = 0$

**Region Represented by  $3x + 4y \geq 12$ :** The line  $3x + 4y = 12$  meets the coordinate axes at  $A(4, 0)$  and  $B(0, 3)$  joining these points by a thick line we get the graph of  $3x + 4y = 12$ . Since  $(0, 0)$  does not satisfy the inequation  $3x + 4y \geq 12$ . So, the portion not containing the origin is represented by the inequation  $3x + 4y \geq 12$ .

**Region Represented by  $y \geq 1$ :** The line  $y = 1$  is parallel to  $x$ -axis at a unit distance from it. Since  $(0, 0)$  does not satisfy the inequation  $y \geq 1$ . So, the region lying above the line  $y = 1$  is represented by  $y \geq 1$ .

**Region Represented by  $x \geq 0$ :** Clearly,  $x \geq 0$  represents the region lying on the right side of  $y$ -axis.

The solution set of the given linear constraints is the intersection of the above regions as shown in Fig. 15.38.

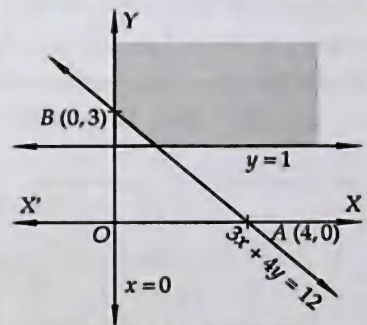


Fig. 15.38

### Type II ON FINDING THE LINEAR INEQUATIONS WHEN THEIR SOLUTION SET IS GIVEN

**EXAMPLE 4** Find the linear inequations for which the shaded area in Fig. 15.39 is the solution set.

**SOLUTION** Consider the line  $x + 2y = 8$ . We observe that the shaded region and the origin are on the same side of the line  $x + 2y = 8$  and  $(0, 0)$  satisfies the linear constraint  $x + 2y \leq 8$ . So, we must have one inequations as  $x + 2y \leq 8$ .

Now consider the line  $2x + y = 2$ . We find that the shaded region and the origin are on the opposite sides of the line  $2x + y = 2$  and  $(0, 0)$  does not satisfy the inequation  $2x + y \geq 2$ . So, the second inequations is  $2x + y \geq 2$ .

Finally, consider the line  $x - y = 1$ . We observe that the shaded region and the origin are on the same side of the line  $x - y = 1$ . We observe that the shaded region and the origin are on the same side of the line  $x - y = 1$  and  $(0, 0)$  satisfies  $x - y \leq 1$ . So, the third constraint is  $x - y \leq 1$ .

We also notice that the shaded region is above  $x$ -axis and is on the right side of  $y$ -axis. So, we must have  $x \geq 0$  and  $y \geq 0$ .

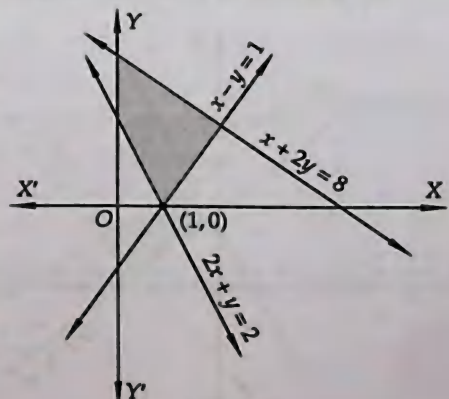


Fig. 15.39

Thus, the linear inequations corresponding to the given solution set are

$$x + 2y \leq 8, 2x + y \geq 2, x - y \leq 1, x \geq 0, y \geq 0$$

**EXAMPLE 5** Find the linear inequations for which the shaded region in Fig. 15.40 is the solution set.

**SOLUTION** Consider the line  $2x + 3y = 3$ . We observe that the shaded region and the origin lie on the opposite side of this line and  $(0, 0)$  satisfies  $2x + 3y \leq 3$ . Therefore, we must have  $2x + 3y \geq 3$  as the linear inequations corresponding to the line  $2x + 3y = 3$ .

Consider the line  $3x + 4y = 18$ . Clearly, the shaded region and the origin lie on the same side of this line and  $(0, 0)$  satisfies the inequation  $3x + 4y \leq 18$ . So, we must have  $3x + 4y \leq 18$  as the linear inequations corresponding to  $3x + 4y = 18$ .

Consider the line  $x - 6y = 3$ . It is evident from the figure that the origin and the shaded region lie on the same side of this line and  $(0, 0)$  satisfies  $x - 6y \leq 3$ . So,  $x - 6y \leq 3$  is the corresponding inequations.

Consider the line  $-7x + 4y = 14$ . We find that the shaded region and the origin are on the same side of this line and  $(0, 0)$  satisfies the inequations  $-7x + 4y \leq 14$ . So, the corresponding linear inequations is  $-7x + 4y \leq 14$ .

Also, the shaded region is in first quadrant only. So, we must have  $x \geq 0$  and  $y \geq 0$ .

Thus, the linear inequations comprising the given solution set are

$$2x + 3y \geq 3, 3x + 4y \leq 18, -7x + 4y \leq 14, x - 6y \leq 3, x \geq 0, y \geq 0$$

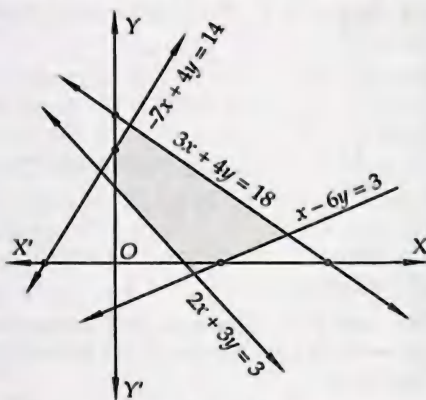


Fig. 15.40

### EXERCISE 15.6

#### LEVEL-1

1. Solve the following systems of linear inequations graphically:

(i)  $2x + 3y \leq 6, 3x + 2y \leq 6, x \geq 0, y \geq 0$  (ii)  $2x + 3y \leq 6, x + 4y \leq 4, x \geq 0, y \geq 0$

(iii)  $x - y \leq 1, x + 2y \leq 8, 2x + y \geq 2, x \geq 0, y \geq 0$

(iv)  $x + y \geq 1, 7x + 9y \leq 63, x \leq 6, y \leq 5, x \geq 0, y \geq 0$

(v)  $2x + 3y \leq 35, y \geq 3, x \geq 2, x \geq 0, y \geq 0$

2. Show that the solution set of the following linear inequations is empty set:

(i)  $x - 2y \geq 0, 2x - y \leq -2, x \geq 0, y \geq 0$  (ii)  $x + 2y \leq 3, 3x + 4y \geq 12, y \geq 1, x \geq 0, y \geq 0$

3. Find the linear inequations for which the shaded area in Fig. 15.41 is the solution set. Draw the diagram of the solution set of the linear inequations:

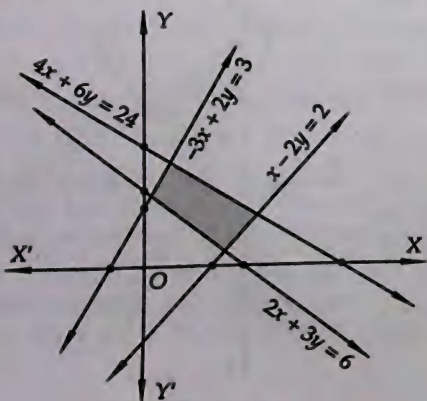


Fig. 15.41

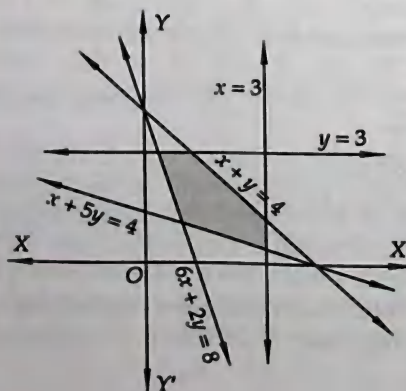


Fig. 15.42

4. Find the linear inequations for which the solution set is the shaded region given in Fig. 15.42.
5. Show that the solution set of the following linear in equations is an unbounded set:  
 $x + y \geq 9$ ,  $3x + y \geq 12$ ,  $x \geq 0$ ,  $y \geq 0$ .
6. Solve the following systems of inequations graphically:
  - (i)  $2x + y \geq 8$ ,  $x + 2y \geq 8$ ,  $x + y \leq 6$
  - (ii)  $12x + 12y \leq 840$ ,  $3x + 6y \leq 300$ ,  $8x + 4y \leq 480$   $x \geq 0$ ,  $y \geq 0$
  - (iii)  $x + 2y \leq 40$ ,  $3x + y \geq 30$ ,  $4x + 3y \geq 60$ ,  $x \geq 0$ ,  $y \geq 0$
  - (iv)  $5x + y \geq 10$ ,  $2x + 2y \geq 12$ ,  $x + 4y \geq 12$ ,  $x \geq 0$ ,  $y \geq 0$
7. Show that the following system of linear equations has no solution:  
 $x + 2y \leq 3$ ,  $3x + 4y \geq 12$ ,  $x \geq 0$ ,  $y \geq 1$ .
8. Show that the solution set of the following system of linear inequalities is an unbounded region  $2x + y \geq 8$ ,  $x + 2y \geq 10$ ,  $x \geq 0$ ,  $y \geq 0$ .

**ANSWERS**

3.  $2x + 3y \geq 6$ ,  $4x + 6y \leq 24$ ,  $-3x + 2y \leq 3$ ,  $x - 2y \leq 2$ ,  $x \geq 0$ ,  $y \geq 0$ ,
4.  $x + y \leq 4$ ,  $y \leq 3$ ,  $x \leq 3$ ,  $x + 5y \geq 4$ ,  $6x + 2y \geq 8$ ,  $x \geq 0$ ,  $y \geq 0$

**VERY SHORT ANSWER QUESTIONS (VSAQs)**

Answer each of the following questions in one word or one sentence or as per exact requirement of the question:

1. Write the solution set of the inequation  $\frac{x^2}{x-2} > 0$ .
2. Write the solution set of the inequation  $x + \frac{1}{x} \geq 2$ .
3. Write the set of values of  $x$  satisfying the inequation  $(x^2 - 2x + 1)(x - 4) \geq 0$ .
4. Write the solution set of the equation  $|2 - x| = x - 2$ .
5. Write the set of values of  $x$  satisfying  $|x - 1| \leq 3$  and  $|x - 1| \leq 1$ .
6. Write the solution set of the inequation  $\left| \frac{1}{x} - 2 \right| < 4$ .
7. Write the number of integral solutions of  $\frac{x+2}{x^2+1} > \frac{1}{2}$ .
8. Write the set of values of  $x$  satisfying the inequations  $5x + 2 < 3x + 8$  and  $\frac{x+2}{x-1} < 4$ .
9. Write the solution set of  $\left| x + \frac{1}{x} \right| > 2$ .
10. Write the solution set of the inequation  $|x - 1| \geq |x - 3|$ .

**ANSWERS**

- |                               |                                         |                   |                  |
|-------------------------------|-----------------------------------------|-------------------|------------------|
| 1. $[2, \infty)$              | 2. $(0, \infty)$                        | 3. $(-\infty, 4)$ | 4. $(2, \infty)$ |
| 5. $[2, 4]$                   | 6. $(-\infty, -1/2) \cup (1/6, \infty)$ | 7. 3              |                  |
| 8. $(-\infty, 1) \cup (2, 3)$ | 9. $R - \{-1, 0, 1\}$                   | 10. $[2, \infty)$ |                  |

**MULTIPLE CHOICE QUESTIONS (MCQs)**

Mark the correct alternative in each of the following:

1. If  $x < 7$ , then
 

|               |                  |               |                  |
|---------------|------------------|---------------|------------------|
| (a) $-x < -7$ | (b) $-x \leq -7$ | (c) $-x > -7$ | (d) $-x \geq -7$ |
|---------------|------------------|---------------|------------------|

2. If  $-3x + 17 < -13$ , then  
 (a)  $x \in (10, \infty)$  (b)  $x \in [10, \infty)$  (c)  $x \in (-\infty, 10]$  (d)  $x \in [-10, 10]$
3. Given that  $x, y$  and  $b$  are real numbers and  $x < y, b > 0$ , then  
 (a)  $\frac{x}{b} < \frac{y}{b}$  (b)  $\frac{x}{b} \leq \frac{y}{b}$  (c)  $\frac{x}{b} > \frac{y}{b}$  (d)  $\frac{x}{b} \geq \frac{y}{b}$
4. If  $x$  is a real number and  $|x| < 5$ , then  
 (a)  $x \geq 5$  (b)  $-5 < x < 5$  (c)  $x \leq -5$  (d)  $-5 \leq x \leq 5$
5. If  $x$  and  $a$  are real numbers such that  $a > 0$  and  $|x| > a$ , then  
 (a)  $x \in (-a, \infty)$  (b)  $x \in [-\infty, a]$  (c)  $x \in (-a, a)$  (d)  $x \in (-\infty, -a) \cup (a, \infty)$
6. If  $|x - 1| > 5$ , then  
 (a)  $x \in (-4, 6)$  (b)  $x \in [-4, 6]$   
 (c)  $x \in (-\infty, -4) \cup (6, \infty)$  (d)  $x \in (-\infty, -4) \cup [6, \infty)$
7. If  $|x + 2| \leq 9$ , then  
 (a)  $x \in (-7, 11)$  (b)  $x \in [-11, 7]$   
 (c)  $x \in (-\infty, -7) \cup (11, \infty)$  (d)  $x \in (-\infty, -7) \cup [11, \infty)$
8. The inequality representing the following graph is  
 (a)  $|x| < 3$  (b)  $|x| \leq 3$  (c)  $|x| > 3$  (d)  $|x| \geq 3$

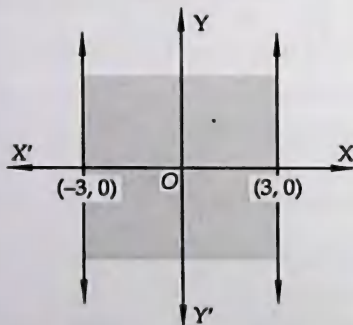


Fig. 15.43

9. The linear inequality representing the solution set given in Fig. 15.44 is  
 (a)  $|x| < 5$  (b)  $|x| > 5$  (c)  $|x| \geq 5$  (d)  $|x| \leq 5$



Fig. 15.44

10. The solution set of the inequation  $|x + 2| \leq 5$  is  
 (a)  $(-7, 5)$  (b)  $[-7, 3]$  (c)  $[-5, 5]$  (d)  $(-7, 3)$
11. If  $\frac{|x-2|}{x-2} \geq 0$ , then  
 (a)  $x \in [2, \infty)$  (b)  $x \in (2, \infty)$  (c)  $x \in (-\infty, 2)$  (d)  $x \in (-\infty, 2]$
12. If  $|x + 3| \geq 10$ , then  
 (a)  $x \in (-13, 7]$  (b)  $x \in (-13, 7)$   
 (c)  $x \in (-\infty, -13) \cup (7, \infty)$  (d)  $x \in (-\infty, -13] \cup [7, \infty)$

**ANSWERS**

1. (c) 2. (a) 3. (a) 4. (b) 5. (d) 6. (c) 7. (b) 8. (b)  
 9. (c) 10. (b) 11. (b) 12. (d)

## PERMUTATIONS

### 16.1 THE FACTORIAL

In this section, we shall introduce the term and notation of factorial which will be often used in this chapter and the next three chapters.

**FACTORIAL** The continued product of first  $n$  natural numbers is called the " $n$  factorial" and is denoted by  $n!$  or  $\lfloor n$ .

i.e.  $n! = 1 \times 2 \times 3 \times 4 \times \dots \times (n-1) \times n$ .

Thus,  $3! = 1 \times 2 \times 3 = 6$ ;  $4! = 1 \times 2 \times 3 \times 4 = 24$ ,  $5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$  etc.

Clearly,  $n!$  is defined for positive integers only.

**ZERO FACTORIAL** As we will require zero factorial in the later sections of this chapter and it does not make any sense to define it as the product of the integers from 1 to zero. So, we define  $0! = 1$ .

**NOTE** Factorials of proper fractions or negative integers are not defined. Factorial  $n$  is defined only for whole numbers.

**DEDUCTION** We have,

$$n! = 1 \times 2 \times 3 \times 4 \dots \times (n-1) \times n = [1 \times 2 \times 3 \times 4 \dots \times (n-1)] n = [(n-1)!] n = n \times (n-1)!$$

Thus,  $n! = n \times (n-1)!$

Similarly,

$$n! = n(n-1)(n-2)! = n(n-1)(n-2)(n-3)! = n(n-1)(n-2)(n-3)(n-4)! \text{ and so on.}$$

For example,  $8! = 8(7!)$ ,  $5! = 5(4!)$  and  $2! = 2(1!)$

Following examples will illustrate the use of this property of factorial  $n$ .

### ILLUSTRATIVE EXAMPLES

#### LEVEL-1

**EXAMPLE 1** Compute: (i)  $\frac{20!}{18!}$  (ii)  $\frac{10!}{6!4!}$

**SOLUTION** (i) We have,

$$\begin{aligned} \frac{20!}{18!} &= \frac{20(19!)}{18!} = \frac{20 \times 19 \times 18!}{18!} \\ &= 20 \times 19 = 380 \end{aligned}$$

$$[\because n! = n \times (n-1)!]$$

$$(ii) \quad \frac{10!}{6!4!} = \frac{10 \times 9 \times 8 \times 7 \times 6!}{6! \times (4 \times 3 \times 2 \times 1)} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210$$

**EXAMPLE 2** Convert the following products into factorials:

$$(i) 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10$$

$$(ii) 2 \cdot 4 \cdot 6 \cdot 8 \cdot 10$$

$$\text{SOLUTION (i) } 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10 = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} = \frac{10!}{5!}$$

$$(ii) \quad 2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 = (2 \times 1)(2 \times 2)(2 \times 3)(2 \times 4)(2 \times 5) = 2^5 \times (1 \cdot 2 \cdot 3 \cdot 4 \cdot 5) = 2^5 \times 5!$$

**EXAMPLE 3** Find the LCM of  $4!$ ,  $5!$  and  $6!$

**SOLUTION** We have,  $5! = 5 \times 4!$  and  $6! = 6 \times 5 \times 4!$

$\therefore$  L.C.M. of  $4!$ ,  $5!$ ,  $6!$  = L.C.M.  $\{4!, 5 \times 4!, 6 \times 5 \times 4!\} = (4!) \times 5 \times 6 = 6! = 720$

**EXAMPLE 4** If  $\frac{1}{9!} + \frac{1}{10!} = \frac{x}{11!}$ , find  $x$ .

**SOLUTION** We have,

$$\frac{1}{9!} + \frac{1}{10!} = \frac{x}{11!}$$

$$\Rightarrow \frac{1}{9!} + \frac{1}{10 \times 9!} = \frac{x}{11 \times 10 \times 9!}$$

$$\Rightarrow \frac{1}{9!} \left( 1 + \frac{1}{10} \right) = \left( \frac{x}{11 \times 10} \right) \times \frac{1}{9!}$$

$$\Rightarrow 1 + \frac{1}{10} = \frac{x}{11 \times 10} \Rightarrow \frac{11}{10} = \frac{x}{11 \times 10} \Rightarrow x = 11 \times 11 = 121.$$

**ALITER** We have,

$$\frac{1}{9!} + \frac{1}{10!} = \frac{x}{11!}$$

Multiplying both sides by the LCM of  $9!$ ,  $10!$  and  $11!$  i.e. by  $11!$ , we obtain

$$\frac{11!}{9!} + \frac{11!}{10!} = \frac{x}{11!} \times 11!$$

$$\Rightarrow \frac{11 \times 10 \times 9!}{9!} + \frac{11 \times 10!}{10!} = x$$

$$\Rightarrow 11 \times 10 + 11 = x$$

$$\Rightarrow x = 121.$$

**EXAMPLE 5** Find  $n$ , if:

$$(i) (n+2)! = 2550 \times n!$$

$$(ii) (n+1)! = 12 \times (n-1)!$$

**SOLUTION** (i) We have,

$$(n+2)! = 2550 \times n!$$

$$\Rightarrow (n+2)(n+1) \times n! = 2550 \times n!$$

$$\Rightarrow (n+2) \times (n+1) = 2550$$

$$\Rightarrow (n+2)(n+1) = 51 \times 50$$

$$\Rightarrow n+2 = 51 \text{ or } n+1 = 50$$

$$\Rightarrow n = 49$$

(ii) We have,

$$(n+1)! = 12 \times (n-1)!$$

$$\Rightarrow (n+1) \times n \times (n-1)! = 12 \times (n-1)!$$

$$\Rightarrow n(n+1) = 12$$

$$\Rightarrow (n+1)n = 4 \times 3 \Rightarrow n = 3$$

[Expressing 2550 as the product of two consecutive natural numbers]

[By comparing]

[By comparing]

**EXAMPLE 6** If  $\frac{n!}{2!(n-2)!}$  and  $\frac{n!}{4!(n-4)!}$  are in the ratio 2 : 1, find the value of  $n$ .

**SOLUTION** We have,

$$\begin{aligned} \frac{n!}{2!(n-2)!} : \frac{n!}{4!(n-4)!} &= 2 : 1 \\ \Rightarrow \frac{n!}{2!(n-2)!} \times \frac{4!(n-4)!}{n!} &= \frac{2}{1} \\ \Rightarrow \frac{4!(n-4)!}{2!(n-2) \times (n-3) \times (n-4)!} &= \frac{2}{1} \\ \Rightarrow \frac{4 \times 3 \times 2!}{2!(n-2)(n-3)} &= \frac{2}{1} \\ \Rightarrow (n-2)(n-3) &= 6 \Rightarrow (n-2)(n-3) = 3 \times 2 \Rightarrow n-2 = 3 \text{ and } n-3 = 2 \Rightarrow n = 5 \end{aligned}$$

**EXAMPLE 7** Prove that:  $\frac{(2n)!}{n!} = \{1 \cdot 3 \cdot 5 \dots (2n-1)\} 2^n$ .

**SOLUTION** We have,

$$\begin{aligned} \frac{(2n)!}{n!} &= \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \dots (2n-2)(2n-1)(2n)}{n!} \\ &= \frac{\{1 \cdot 3 \cdot 5 \cdot 7 \dots (2n-1)\} \cdot \{2 \cdot 4 \cdot 6 \cdot 8 \dots (2n-2)(2n)\}}{n!} \\ &= \frac{\{1 \cdot 3 \cdot 5 \cdot 7 \dots (2n-1)\} 2^n \{1 \cdot 2 \cdot 3 \cdot 4 \dots (n-1)n\}}{n!} \\ &= \frac{\{1 \cdot 3 \cdot 5 \cdot 7 \dots (2n-1)\} \cdot 2^n \cdot n!}{n!} = \{1 \cdot 3 \cdot 5 \cdot 7 \dots (2n-1)\} 2^n \end{aligned}$$

**LEVEL-2**

**EXAMPLE 8** Prove that  $(n! + 1)$  is not divisible by any natural number between 2 and  $n$ .

**SOLUTION** Let  $m$  be divisible by  $k$  and  $r$  be any natural number between 1 and  $k$ . If  $m + r$  is divided by  $k$ , then we obtain  $r$  as the remainder.

We have,  $n! = 1 \cdot 2 \cdot 3 \cdot 4 \dots (n-1) \cdot n$ .

Therefore,  $n!$  is divisible by every natural number between 2 and  $n$ . So,  $(n! + 1)$ , when divided by any natural number between 2 and  $n$ , leaves 1 as the remainder.

Hence,  $(n! + 1)$  is not divisible by any natural number between 2 and  $n$ .

**EXAMPLE 9** Prove the inequalities  $(n!)^2 \leq n^n (n!) < (2n)!$  for all positive integers  $n$ .

**SOLUTION** Clearly,

$$(n!)^2 = (n!)(n!) = (1 \cdot 2 \cdot 3 \cdot 4 \dots (n-1)n)(n!)$$

$$\begin{aligned} \text{Now, } \left. \begin{array}{l} 1 \leq n \\ 2 \leq n \\ 3 \leq n \\ \vdots \\ (n-1) \leq n \\ n \leq n \end{array} \right\} &\Rightarrow 1 \cdot 2 \cdot (n-1)n \leq \underbrace{n \cdot n \cdot n \dots n}_{n\text{-times}} \end{aligned}$$

$$\Rightarrow n! \leq n^n \Rightarrow (n!)(n!) \leq n^n (n!) \Rightarrow (n!)^2 \leq n^n (n!) \quad \dots(i)$$

We have,  $(2n)! = 1 \cdot 2 \cdot 3 \dots (n-1)n(n+1)(n+2) \dots (2n-1)(2n)$

$$\Rightarrow (2n)! = n!(n+1)(n+2)\dots(2n-1)(2n)$$

$$\text{Now, } \left. \begin{array}{l} n+1 > n \\ n+2 > n \\ n+3 > n \\ \vdots \\ n+(n-1) > n \\ n+n > n \end{array} \right\} \Rightarrow (n+1)(n+2)(n+3)\dots(2n-1)(2n) > n^n$$

$$\Rightarrow n!(n+1)(n+2)\dots(2n-1)(2n) > n!n^n$$

$$\Rightarrow (2n)! > n!n^n \Rightarrow n!n^n < (2n)! \quad \dots(ii)$$

From (i) and (ii), we get  $(n!)^2 \leq n^n (n!) < (2n)!$

**EXAMPLE 10** Prove that  $33!$  is divisible by  $2^{15}$ . What is the largest integer  $n$  such that  $33!$  is divisible by  $2^n$ ?

**SOLUTION** Let  $E_2(n)$  denote the index of 2 in  $n$ . Then,

$$E_2(33!) = E_2(1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \dots 32 \cdot 33)$$

$$\Rightarrow E_2(33!) = E_2(2 \cdot 4 \cdot 6 \cdot 8 \dots 30 \cdot 32)$$

$$\Rightarrow E_2(33!) = 16 + E_2(1 \cdot 2 \cdot 3 \dots 15 \cdot 16)$$

$$\Rightarrow E_2(33!) = 16 + E_2(2 \cdot 4 \cdot 6 \dots 14 \cdot 16)$$

$$\Rightarrow E_2(33!) = 16 + 8 + E_2(1 \cdot 2 \cdot 3 \dots 8)$$

$$\Rightarrow E_2(33!) = 16 + 8 + E_2(2 \cdot 4 \cdot 6 \cdot 8)$$

$$\Rightarrow E_2(33!) = 16 + 8 + 4 + E_2(1 \cdot 2 \cdot 3 \cdot 4)$$

$$\Rightarrow E_2(33!) = 16 + 8 + 4 + E_2(2 \cdot 4) = 16 + 8 + 4 + 3 = 31.$$

Thus, exponent of 2 in  $33!$  is 31 i.e.  $33! = 2^{31} \times \text{an integer}$

This shows that  $33!$  is divisible by  $2^{15}$  and the largest integer  $n$  such that  $33!$  is divisible by  $2^n$  is 31.

### EXERCISE 16.1

#### LEVEL-1

1. Compute:

(i)  $\frac{30!}{28!}$

(ii)  $\frac{11! - 10!}{9!}$

(iii) L.C.M.  $(6!, 7!, 8!)$

2. Prove that  $\frac{1}{9!} + \frac{1}{10!} + \frac{1}{11!} = \frac{122}{11!}$

3. Find  $x$  in each of the following:

(i)  $\frac{1}{4!} + \frac{1}{5!} = \frac{x}{6!}$

(ii)  $\frac{x}{10!} = \frac{1}{8!} + \frac{1}{9!}$

[NCERT]

(iii)  $\frac{1}{6!} + \frac{1}{7!} = \frac{x}{8!}$

[NCERT]

4. Convert the following products into factorials:

(i)  $5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10$

(ii)  $3 \cdot 6 \cdot 9 \cdot 12 \cdot 15 \cdot 18$

(iii)  $(n+1)(n+2)(n+3)\dots(2n)$

(iv)  $1 \cdot 3 \cdot 5 \cdot 7 \cdot 9 \dots (2n-1)$

5. Which of the following are true:

(i)  $(2+3)! = 2! + 3!$

(ii)  $(2 \times 3)! = 2! \times 3!$

6. Prove that  $n!(n+2) = n! + (n+1)!$

7. If  $(n+2)! = 60[(n-1)!]$ , find  $n$ .

8. If  $(n+1)! = 90[(n-1)!]$ , find  $n$ .